

Determinants

Chapter 4 · All Questions and Solutions



Part 1 · Examples

Q

1

Evaluating a
2x2
determinant



Evaluate $\begin{vmatrix} 2 & 4 \\ -1 & 2 \end{vmatrix}$.

Conclusion

We have $\begin{vmatrix} 2 & 4 \\ -1 & 2 \end{vmatrix} = 2(2) - 4(-1) = 4 + 4 = 8$.

Q

2

Evaluating a
2x2
determinant
with
variables



Evaluate $\begin{vmatrix} x & x+1 \\ x-1 & x \end{vmatrix}$

Conclusion

We have

$$\begin{vmatrix} x & x+1 \\ x-1 & x \end{vmatrix} = x(x) - (x+1)(x-1) = x^2 - (x^2 - 1) = x^2 - x^2 + 1 = 1$$

Q

3

Evaluating a
3x3
determinant

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Evaluate the determinant $\Delta = \begin{vmatrix} 1 & 2 & 4 \\ -1 & 3 & 0 \\ 4 & 1 & 0 \end{vmatrix}$.

Given

Note that in the third column, two entries are zero.

So we expand along the third column (C_3).

We get

Key formula

$$\Delta = 4 \begin{vmatrix} -1 & 3 \\ 4 & 1 \end{vmatrix} - 0 \begin{vmatrix} 1 & 2 \\ 4 & 1 \end{vmatrix} + 0 \begin{vmatrix} 1 & 2 \\ -1 & 3 \end{vmatrix} = 4(-1-12) - 0 + 0 = -52$$

Q

4

Evaluating a
3x3
determinant
with trig
entries



Evaluate $\Delta = \begin{vmatrix} 0 & \sin \alpha & -\cos \alpha \\ -\sin \alpha & 0 & \sin \beta \\ \cos \alpha & -\sin \beta & 0 \end{vmatrix}$.

Given

We expand along R_1 .

We get

Key formula

$$\Delta = 0 \begin{vmatrix} 0 & \sin \beta \\ -\sin \beta & 0 \end{vmatrix} \begin{vmatrix} -\sin \alpha \\ \cos \alpha \end{vmatrix} \begin{vmatrix} \sin \beta \\ 0 \end{vmatrix} \begin{vmatrix} -\sin \alpha & \sin \beta \\ \cos \alpha & 0 \end{vmatrix} \begin{vmatrix} -\cos \alpha \\ \cos \alpha \end{vmatrix} \begin{vmatrix} -\sin \alpha & 0 \\ \cos \alpha & -\sin \beta \end{vmatrix}$$

$$= 0 - \sin \alpha (0 - \sin \beta \cos \alpha) - \cos \alpha (\sin \alpha \sin \beta - 0)$$

$$= \sin \alpha \sin \beta \cos \alpha - \cos \alpha \sin \alpha \sin \beta = 0$$

Q**5**Solving for x
from equal
determinantsFind values of x for which $\begin{vmatrix} 3 & x \\ x & 1 \end{vmatrix} = \begin{vmatrix} 3 & 2 \\ 4 & 1 \end{vmatrix}$.**Substitute**

We have $\begin{vmatrix} 3 & x \\ x & 1 \end{vmatrix} = \begin{vmatrix} 3 & 2 \\ 4 & 1 \end{vmatrix}$.

That is,

$$3 - x^2 = 3 - 8$$

That is,

$$x^2 = 8$$

Conclusion

So,

$$x = \pm 2\sqrt{2}$$

Q**6**Area of a
triangle
using
determinants

Find the area of the triangle whose vertices are (3, 8), (-4, 2) and (5, 1).

Given

The area of the triangle is given by

Key formula

$$\Delta = \frac{1}{2} \begin{vmatrix} 3 & 8 & 1 \\ -4 & 2 & 1 \\ 5 & 1 & 1 \end{vmatrix}$$

Conclusion

$$= \frac{1}{2} [3(2-1) - 8(-4-5) + 1(-4-10)]$$

$$= \frac{1}{2} (3 + 72 - 14) = \frac{61}{2}$$

Q**7**Equation of a
line via
determinants**(())**

Find the equation of the line joining A(1,3) and B(0,0) using determinants and find k if D(k, 0) is a point such that area of triangle ABD is 3sq units.

Given

Let P(x, y) be any point on AB.

Then the area of triangle ABP is zero (Why?).

So

$$\frac{1}{2} \begin{vmatrix} 0 & 0 & 1 \\ 1 & 3 & 1 \\ x & y & 1 \end{vmatrix} = 0$$

This gives

$$\frac{1}{2} (y-3x) = 0 \text{ or } y = 3x$$

This is the equation of the required line AB.

Conclusion

Also, since the area of the triangle ABD is 3 sq. units, we have

$$\frac{1}{2} \begin{vmatrix} 1 & 3 & 1 \\ 0 & 0 & 1 \\ k & 0 & 1 \end{vmatrix} = \pm 3$$

This gives $\frac{-3k}{2} = \pm 3$, that is, $k = \mp 2$.

Q

8

Minor of a determinant element



Find the minor of element 6 in the determinant $\Delta = \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix}$

Given

Since 6 lies in the second row and third column, its minor M_{23} is given by

Key formula

$$M_{23} = \begin{vmatrix} 1 & 2 \\ 7 & 8 \end{vmatrix} = 8 - 14 = -6 \text{ (obtained by deleting } R_2 \text{ and } C_3 \text{ in } \Delta).$$

Given

Definition 2: The cofactor of an element a_{ij} , denoted by A_{ij} , is defined by

$$A_{ij} = (-1)^{i+j} M_{ij}, \text{ where } M_{ij} \text{ is minor of } a_{ij}.$$

Q

9

Minors and cofactors of a 2x2 determinant



Find minors and cofactors of all the elements of the determinant $\begin{vmatrix} 1 & -2 \\ 4 & 3 \end{vmatrix}$

Given

Minor of the element a_{ij} is M_{ij}

Here $a_{11} = 1$.

So $M_{11} = \text{Minor of } a_{11} = 3$

$M_{12} = \text{Minor of the element } a_{12} = 4$

$M_{21} = \text{Minor of the element } a_{21} = -2$

$M_{22} = \text{Minor of the element } a_{22} = 1$

Now, cofactor of a_{ij} is A_{ij} .

So

Key formula

$$A_{11} = (-1)^{1+1} M_{11} = (-1)^2(3) = 3$$

$$A_{12} = (-1)^{1+2} M_{12} = (-1)^3(4) = -4$$

$$A_{21} = (-1)^{2+1} M_{21} = (-1)^3(-2) = 2$$

$$A_{22} = (-1)^{2+2} M_{22} = (-1)^4(1) = 1$$

Q

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Minors and
cofactors of a
3x3
determinant



Find minors and cofactors of the elements a_{11} , a_{21} in the determinant

$$\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

Given

By definition of minors and cofactors, we have

Key formula

$$\text{Minor of } a_{11} = M_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} = a_{22} a_{33} - a_{23} a_{32}$$

Conclusion

$$\text{Cofactor of } a_{11} = A_{11} = (-1)^{1+1} M_{11} = a_{22} a_{33} - a_{23} a_{32}$$

$$\text{Minor of } a_{21} = M_{21} = \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix} = a_{12} a_{33} - a_{13} a_{32}$$

$$\text{Cofactor of } a_{21} = A_{21} = (-1)^{2+1} M_{21} = (-1)(a_{12} a_{33} - a_{13} a_{32}) = -a_{12} a_{33} + a_{13} a_{32}$$

Cofactor expansion

Remark: We expand the determinant Δ in Example 21, along R_1 .

We get

$$\begin{aligned} \Delta &= (-1)^{1+1} a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} + (-1)^{1+2} a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + (-1)^{1+3} a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} \\ &= a_{11} A_{11} + a_{12} A_{12} + a_{13} A_{13}, \text{ where } A_{ij} \text{ is cofactor of } a_{ij} \\ &= \text{sum of product of elements of } R_1 \text{ with their corresponding cofactors} \end{aligned}$$

Similarly, Δ can be calculated in five other ways, by expanding along R_2 , R_3 , C_1 , C_2 and C_3 .

So, Δ = is the sum of the product of elements of any row (or column) with their corresponding cofactors.

Zero-sum property

- Note: If elements of a row (or column) are multiplied with cofactors of any other row (or column), then their sum is zero.

For example,

$$\begin{aligned} \Delta &= a_{11} A_{21} + a_{12} A_{22} + a_{13} A_{23} \\ &= a_{11} (-1)^{1+1} \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix} + a_{12} (-1)^{1+2} \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} (-1)^{1+3} \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix} \\ &= \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{11} & a_{12} & a_{13} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = 0 \text{ (since } R_1 \text{ and } R_2 \text{ are identical)} \end{aligned}$$

Similarly, we can try for other rows and columns.

Q

11

Verifying a
cofactor
expansion
identity

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Find minors and cofactors of the elements of the determinant

$$\begin{vmatrix} 2 & -3 & 5 \\ 6 & 0 & 4 \\ 1 & 5 & -7 \end{vmatrix} \text{ and}$$

verify that $a_{11} A_{31} + a_{12} A_{32} + a_{13} A_{33} = 0$

Given

We have $M_{11} = \begin{vmatrix} 0 & 4 \\ 5 & -7 \end{vmatrix} = 0 \cdot (-7) - 20 = -20$; $A_{11} = (-1)^{1+1}(-20) = -20$

$$M_{12} = \begin{vmatrix} 6 & 4 \\ 1 & -7 \end{vmatrix} = -42 - 4 = -46$$
; $A_{12} = (-1)^{1+2}(-46) = 46$

$$M_{13} = \begin{vmatrix} 6 & 0 \\ 1 & 5 \end{vmatrix} = 30 - 0 = 30$$
; $A_{13} = (-1)^{1+3}(30) = 30$

$$M_{21} = \begin{vmatrix} -3 & 5 \\ 5 & -7 \end{vmatrix} = 21 - 25 = -4$$
; $A_{21} = (-1)^{2+1}(-4) = 4$

$$M_{22} = \begin{vmatrix} 2 & 5 \\ 1 & -7 \end{vmatrix} = -14 - 5 = -19$$
; $A_{22} = (-1)^{2+2}(-19) = -19$

$$M_{23} = \begin{vmatrix} 2 & -3 \\ 1 & 5 \end{vmatrix} = 10 + 3 = 13$$
; $A_{23} = (-1)^{2+3}(13) = -13$

$$M_{31} = \begin{vmatrix} -3 & 5 \\ 0 & 4 \end{vmatrix} = -12 - 0 = -12$$
; $A_{31} = (-1)^{3+1}(-12) = -12$

$$M_{32} = \begin{vmatrix} 2 & 5 \\ 6 & 4 \end{vmatrix} = 8 - 30 = -22$$
; $A_{32} = (-1)^{3+2}(-22) = 22$

and

$$M_{33} = \begin{vmatrix} 2 & -3 \\ 6 & 0 \end{vmatrix} = 0 + 18 = 18$$
; $A_{33} = (-1)^{3+3}(18) = 18$

Conclusion

Now

$$a_{11} = 2, a_{12} = -3, a_{13} = 5; A_{31} = -12, A_{32} = 22, A_{33} = 18$$

So

$$\begin{aligned} & a_{11} A_{31} + a_{12} A_{32} + a_{13} A_{33} \\ &= 2(-12) + (-3)(22) + 5(18) = -24 - 66 + 90 = 0 \end{aligned}$$

Q

12

Finding the
adjoint of a
matrix



Find adj A for $A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$

Given

We have $A_{11} = 4, A_{12} = -1, A_{21} = -3, A_{22} = 2$

Hence

Key formula

$$\text{adj } A = \begin{bmatrix} A_{11} & A_{21} \\ A_{12} & A_{22} \end{bmatrix} = \begin{bmatrix} 4 & -3 \\ -1 & 2 \end{bmatrix}$$

Q

13

Verifying
 $A(\text{adj } A) = |A| I$



If $A = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$, then verify that $A \text{adj } A = |A| I$. Also find A^{-1} .

Given

We have $|A| = 1(16-9)-3(4-3)+3(3-4) = 1 \neq 0$

Now $A_{11} = 7, A_{12} = -1, A_{13} = -1, A_{21} = -3, A_{22} = 1, A_{23} = 0, A_{31} = -3, A_{32} = 0, A_{33} = 1$

Substitute

Therefore

Key formula

$$\text{adj } A = \begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

Substitute

Now

$$\begin{aligned} A(\text{adj } A) &= \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix} \begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 7-3-3 & -3+3+0 & -3+0+3 \\ 7-4-3 & -3+4+0 & -3+0+3 \\ 7-3-4 & -3+3+0 & -3+0+4 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = (1) \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = |A| \cdot I \end{aligned}$$

Conclusion

Also

$$A^{-1} = \frac{1}{|A|} \text{adj } A = \frac{1}{1} \begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

Q

14

Verifying
 $(AB)^{-1} = B^{-1} A^{-1}$



If $A = \begin{bmatrix} 2 & 3 \\ 1 & -4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix}$, then verify that $(AB)^{-1} = B^{-1} A^{-1}$.

Given

$$\text{We have } AB = \begin{bmatrix} 2 & 3 \\ 1 & -4 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} -1 & 5 \\ 5 & -14 \end{bmatrix}$$

Since $|AB| = -11 \neq 0$, $(AB)^{-1}$ exists, it is given by

$$(AB)^{-1} = \frac{1}{|AB|} \text{adj}(AB) = -\frac{1}{11} \begin{bmatrix} -14 & -5 \\ -5 & -1 \end{bmatrix} = \frac{1}{11} \begin{bmatrix} 14 & 5 \\ 5 & 1 \end{bmatrix}$$

Substitute

Also, $|A| = -11 \neq 0$ and $|B| = 1 \neq 0$.

So A^{-1} and B^{-1} both exist.

They are given by

$$A^{-1} = -\frac{1}{11} \begin{bmatrix} -4 & -3 \\ -1 & 2 \end{bmatrix}, B^{-1} = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$$

Conclusion

Therefore

$$B^{-1} A^{-1} = -\frac{1}{11} \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -4 & -3 \\ -1 & 2 \end{bmatrix} = -\frac{1}{11} \begin{bmatrix} -14 & -5 \\ -5 & -1 \end{bmatrix} = \frac{1}{11} \begin{bmatrix} 14 & 5 \\ 5 & 1 \end{bmatrix}$$

$$\text{Hence } (AB)^{-1} = B^{-1} A^{-1}$$

Q

15

Matrix polynomial identity and inverse via it

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Show that the matrix $A = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}$ satisfies the equation $A^2 - 4A + I = O$, where I is 2×2 identity matrix and O is 2×2 zero matrix. Using this equation, find A^{-1} .

Given

$$\text{We have } A^2 = A \cdot A = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 7 & 12 \\ 4 & 7 \end{bmatrix}$$

Hence

$$A^2 - 4A + I = \begin{bmatrix} 7 & 12 \\ 4 & 7 \end{bmatrix} - \begin{bmatrix} 8 & 12 \\ 4 & 8 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O$$

Substitute

Now

$$A^2 - 4A + I = O$$

Therefore

$$AA - 4A = -I$$

or

$$A(A A^{-1}) - 4A A^{-1} = -I A^{-1} \text{ (Post multiplying by } A^{-1} \text{ because } |A| \neq 0)$$

or

$$A(A A^{-1}) - 4I = -A^{-1}$$

or

$$AI - 4I = -A^{-1}$$

or

$$A^{-1} = 4I - A = \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} - \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 2 & -3 \\ -1 & 2 \end{bmatrix}$$

Conclusion

Hence

$$A^{-1} = \begin{bmatrix} 2 & -3 \\ -1 & 2 \end{bmatrix}$$

Q

16

Solving a system of equations by matrix method

(())

Solve the system of equations

$$\begin{aligned} 2x + 5y &= 1 \\ 3x + 2y &= 7 \end{aligned}$$

Given

The system of equations can be written in the form $AX = B$, where

$$A = \begin{bmatrix} 2 & 5 \\ 3 & 2 \end{bmatrix}, X = \begin{bmatrix} x \\ y \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 \\ 7 \end{bmatrix}$$

Substitute

$$\text{Now, } |A| = -11 \neq 0.$$

So A is a nonsingular matrix and has a unique solution.

Note that

$$A^{-1} = -\frac{1}{11} \begin{bmatrix} 2 & -5 \\ -3 & 2 \end{bmatrix}$$

Therefore

$$X = A^{-1} B = -\frac{1}{11} \begin{bmatrix} 2 & -5 & 1 \\ -3 & 2 & 7 \end{bmatrix}$$

i.e.

$$\begin{bmatrix} x \\ y \end{bmatrix} = -\frac{1}{11} \begin{bmatrix} -33 & 3 \\ 11 & -1 \end{bmatrix}$$

Conclusion

Hence

$$x = 3, y = -1$$

Q

17

Equation of a line and finding k via matrix method



Solve the following system of equations by matrix method.

$$\begin{aligned} 3x - 2y + 3z &= 8 \\ 2x + y - z &= 1 \\ 4x - 3y + 2z &= 4 \end{aligned}$$

Given



The system of equations can be written in the form $AX = B$, where

$$A = \begin{bmatrix} 3 & -2 & 3 \\ 2 & 1 & -1 \\ 4 & -3 & 2 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } B = \begin{bmatrix} 8 \\ 1 \\ 4 \end{bmatrix}$$

Substitute

We see that

$$|A| = 3(2-3) + 2(4-4) + 3(-6-4) = -17 \neq 0$$

Hence, A is nonsingular.

So its inverse exists.

Now

$$A_{11} = -1, A_{12} = -8, A_{13} = -10$$

$$A_{21} = -5, A_{22} = -6, A_{23} = 1$$

$$A_{31} = -1, A_{32} = 9, A_{33} = 7$$

Therefore

Key formula



$$A^{-1} = -\frac{1}{17} \begin{bmatrix} -1 & -5 & -1 \\ -8 & -6 & 1 \\ -10 & 9 & 7 \end{bmatrix}$$

Substitute

So

$$X = A^{-1} B = -\frac{1}{17} \begin{bmatrix} -1 & -5 & -1 \\ -8 & -6 & 1 \\ -10 & 9 & 7 \end{bmatrix} \begin{bmatrix} 8 \\ 1 \\ 4 \end{bmatrix}$$

i.e.

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = -\frac{1}{17} \begin{bmatrix} -17 \\ -34 \\ -51 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Conclusion

Hence

$$x = 1, y = 2 \text{ and } z = 3.$$

Q

18

Solving a
word
problem via
matrix
method



The sum of three numbers is 6 . If we multiply third number by 3 and add second number to it, we get 11 . By adding first and third numbers, we get double of the second number. Represent it algebraically and find the numbers using matrix method.

Given

Let the first, second and third numbers be x, y and z, respectively.

Using the given conditions, we get

$$x+y+z = 6$$

$$y+3z = 11$$

$$x+z = 2y \text{ or } x-2y+z = 0$$

This system can be written as $AX = B$, where

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 3 \\ 1 & 2 & 1 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } B = \begin{bmatrix} 6 \\ 11 \\ 0 \end{bmatrix}$$

Substitute

$$\text{Here } |A| = 1(1+6)-(0-3)+(0-1) = 9 \neq 0.$$

Now we find adj A

$$A_{11} = 1(1+6) = 7, A_{12} = -(0-3) = 3, A_{13} = -1$$

$$A_{21} = -(1+2) = -3, A_{22} = 0, A_{23} = -(-2-1) = 3$$

$$A_{31} = (3-1) = 2, A_{32} = -(3-0) = -3, A_{33} = (1-0) = 1$$

Key formula

Hence

$$\text{adj } A = \begin{bmatrix} 7 & -3 & 2 \\ 3 & 0 & -3 \\ -1 & 3 & 1 \end{bmatrix}$$

Thus

$$A^{-1} = \frac{1}{|A|} \text{adj}(A) = \frac{1}{9} \begin{bmatrix} 7 & -3 & 2 \\ 3 & 0 & -3 \\ -1 & 3 & 1 \end{bmatrix}$$

Substitute

Since

$$X = A^{-1} B$$

$$X = \frac{1}{9} \begin{bmatrix} 7 & -3 & 2 \\ 3 & 0 & -3 \\ -1 & 3 & 1 \end{bmatrix} \begin{bmatrix} 6 \\ 11 \\ 0 \end{bmatrix}$$

or

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{9} \begin{bmatrix} 42-33+0 \\ 18+0+0 \\ -6+33+0 \end{bmatrix} = \frac{1}{9} \begin{bmatrix} 9 \\ 18 \\ 27 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Conclusion

Thus

$$x = 1, y = 2, z = 3$$

Q**19**

**Solving a
system using
a product of
matrices**

(())

Use product $\begin{bmatrix} 1 & 1 & 2 \\ 0 & 2 & 3 \\ 3 & 2 & 4 \end{bmatrix} \begin{bmatrix} 2 & 0 & 1 \\ 9 & 2 & 3 \\ 6 & 1 & 2 \end{bmatrix}$ to solve the system of equations

$$\begin{aligned} x-y+2z &= 1 \\ 2y-3z &= 1 \\ 3x-2y+4z &= 2 \end{aligned}$$

Given

Consider the product $\begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & -3 \\ 3 & -2 & 4 \end{bmatrix} \begin{bmatrix} -2 & 0 & 1 \\ 9 & 2 & -3 \\ 6 & 1 & -2 \end{bmatrix}$

$$= \begin{bmatrix} -2-9+12 & 0-2+2 & 1+3-4 \\ 0+18-18 & 0+4-3 & 0-6+6 \\ -6-18+24 & 0-4+4 & 3+6-8 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & -3 \\ 3 & -2 & 4 \end{bmatrix}^{-1} = \begin{bmatrix} -2 & 0 & 1 \\ 9 & 2 & -3 \\ 6 & 1 & -2 \end{bmatrix}$$

Hence

Now, the given system of equations can be written in matrix form as follows

$$\begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & -3 \\ 3 & -2 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

Substitute

or

$$\begin{aligned} \mathbf{x} &= \begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & -3 \\ 3 & -2 & 4 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 1 \\ 9 & 2 & 3 \\ 6 & 1 & 2 \end{bmatrix} \\ &= \begin{bmatrix} -2+0+2 \\ 9+2-6 \\ 6+1-4 \end{bmatrix} = \begin{bmatrix} 0 \\ 5 \\ 3 \end{bmatrix} \end{aligned}$$

Conclusion

Hence

$$x = 0, y = 5 \text{ and } z = 3$$

Q

1

Evaluating a
2x2
determinant



Evaluate the determinants in Exercises 1 and 2.

$$\begin{vmatrix} 2 & 4 \\ -5 & -1 \end{vmatrix}$$

Substitute

$$\text{Let } |A| = \begin{vmatrix} 2 & 4 \\ -5 & -1 \end{vmatrix}$$

Hence,

$$\begin{aligned} |A| &= \begin{vmatrix} 2 & 4 \\ -5 & -1 \end{vmatrix} \\ &= 2(-1) - 4(-5) \\ &= -2 + 20 \\ &= 18 \end{aligned}$$

Q

2

Evaluating
trig 2x2
determinants



Evaluate the determinants in Exercises 1 and 2.

Part (i)

$$\begin{vmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{vmatrix}$$

Part (ii)

$$\begin{vmatrix} x^2 - x + 1 & x - 1 \\ x + 1 & x + 1 \end{vmatrix}$$

Given



(i)

$$\begin{aligned} \begin{vmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{vmatrix} &= (\cos \theta)(\cos \theta) - (-\sin \theta)(\sin \theta) \\ &= \cos^2 \theta + \sin^2 \theta \\ &= 1 \end{aligned}$$

Conclusion

(ii)

$$\begin{aligned} \begin{vmatrix} x^2 - x + 1 & x - 1 \\ x + 1 & x + 1 \end{vmatrix} &= (x^2 - x + 1)(x + 1) - (x - 1)(x + 1) \\ &= x^3 - x^2 + x + x^2 - x + 1 - (x^2 - 1) \\ &= x^3 + 1 - x^2 + 1 \\ &= x^3 - x^2 + 2 \end{aligned}$$

Q

3

Verifying
 $|3A| = 27|A|$
for a 3x3
matrix



If $A = \begin{bmatrix} 1 & 2 \\ 4 & 2 \end{bmatrix}$, then show that $|2A| = 4|A|$

Given



$$\text{The given matrix is } A = \begin{bmatrix} 1 & 2 \\ 4 & 2 \end{bmatrix}$$

Substitute

Therefore,

$$\begin{aligned} 2A &= 2 \begin{pmatrix} 1 & 2 \\ 4 & 2 \end{pmatrix} \\ &= \begin{pmatrix} 2 & 4 \\ 8 & 4 \end{pmatrix} \end{aligned}$$

Hence,

$$\begin{aligned} \text{LHS} &= |2A| \\ &= \begin{vmatrix} 2 & 4 \\ 8 & 4 \end{vmatrix} \\ &= 2 \times 4 - 4 \times 8 \\ &= 8 - 32 \\ &= -24 \end{aligned}$$

Now,

$$\begin{aligned} |A| &= \begin{vmatrix} 1 & 2 \\ 4 & 2 \end{vmatrix} = 1 \times 2 - 2 \times 4 \\ &= 2 - 8 \\ &= -6 \end{aligned}$$

Therefore,

$$\begin{aligned} \text{RHS} &= 4|A| \\ &= 4(-6) \\ &= -24 \end{aligned}$$

Conclusion

Thus, $|2A| = 4|A|$ proved.

Q

4

Evaluating a
3x3
determinant

(())

If $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 4 \end{bmatrix}$, then show that $|3A| = 27|A|$

Given

The given matrix is

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 4 \end{bmatrix}$$

Expansion strategy

In the first column, two entries are zero.

So we expand along the first column (C_1).

This makes the calculation easier.

Substitute

$$\begin{aligned} |A| &= 1 \begin{vmatrix} 1 & 2 \\ 0 & 4 \end{vmatrix} - 0 \begin{vmatrix} 0 & 1 \\ 1 & 4 \end{vmatrix} + 0 \begin{vmatrix} 0 & 1 \\ 1 & 2 \end{vmatrix} \\ &= 1(4-0) - 0 + 0 \\ &= 4 \end{aligned}$$

Therefore,

$$\begin{aligned} 27|A| &= 27|4| \\ &= 108 \end{aligned}$$

Now,

$$3A = 3 \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 3 \\ 0 & 3 & 6 \\ 0 & 0 & 12 \end{bmatrix}$$

Therefore,

$$\begin{aligned} |3A| &= 3 \begin{vmatrix} 3 & 6 \\ 0 & 12 \end{vmatrix} - 0 \begin{vmatrix} 0 & 3 \\ 0 & 12 \end{vmatrix} + 0 \begin{vmatrix} 0 & 3 \\ 3 & 6 \end{vmatrix} \\ &= 3(36-0) \\ &= 36(36) \\ &= 108 \end{aligned}$$

From equations (1) and (2),

$$|3A| = 27|A|$$

Conclusion

Thus, $|3A| = 27|A|$ proved.

Q

5

Evaluating
3x3
determinants



Evaluate the determinants

Part (i)

$$\begin{vmatrix} 3 & -1 & -2 \\ 0 & 0 & -1 \\ 3 & -5 & 0 \end{vmatrix}$$

Part (ii)

$$\begin{vmatrix} 3 & -4 & 5 \\ 1 & 1 & -2 \\ 2 & 3 & 1 \end{vmatrix}$$

Part (iii)

$$\begin{vmatrix} 0 & 1 & 2 \\ -1 & 0 & -3 \\ -2 & 3 & 0 \end{vmatrix}$$

Part (iv)

$$\begin{vmatrix} 2 & -1 & -2 \\ 0 & 2 & -1 \\ 3 & -5 & 0 \end{vmatrix}$$

Given



(i) Let

$$A = \begin{vmatrix} 3 & -1 & -2 \\ 0 & 0 & -1 \\ 3 & -5 & 0 \end{vmatrix}$$

In the second row, two entries are zero.

So we expand along the second row.

This makes the calculation easier.

Hence,

$$\begin{aligned} |A| &= -0 \begin{vmatrix} -1 & -2 \\ -5 & 0 \end{vmatrix} + 0 \begin{vmatrix} 3 & -2 \\ 3 & 0 \end{vmatrix} - (-1) \begin{vmatrix} 3 & -1 \\ 3 & -5 \end{vmatrix} \\ &= (-15+3) \\ &= -12 \end{aligned}$$

Substitute

$$(ii) \text{ Let } A = \begin{vmatrix} 3 & -4 & 5 \\ 1 & 1 & -2 \\ 2 & 3 & 1 \end{vmatrix}$$

Hence,

$$\begin{aligned} |A| &= 3 \begin{vmatrix} 1 & -2 \\ 3 & 1 \end{vmatrix} + 4 \begin{vmatrix} 1 & -2 \\ 2 & 1 \end{vmatrix} + 5 \begin{vmatrix} 1 & 1 \\ 2 & 3 \end{vmatrix} \\ &= 3(1+6) + 4(1+4) + 5(3-2) \\ &= 3(7) + 4(5) + 5(1) \\ &= 21 + 20 + 5 \\ &= 46 \end{aligned}$$

(iii) Let

$$A = \begin{vmatrix} 0 & 1 & 2 \\ -1 & 0 & -3 \\ -2 & 3 & 0 \end{vmatrix}$$

Hence,

$$\begin{aligned}
 |A| &= 0 \begin{vmatrix} 0 & -3 \\ 3 & 0 \end{vmatrix} - 1 \begin{vmatrix} -1 & -3 \\ -2 & 0 \end{vmatrix} + 2 \begin{vmatrix} -1 & 0 \\ -2 & 3 \end{vmatrix} \\
 &= 0 - 1(0 - 6) + 2(-3 - 0) \\
 &= -1(-6) + 2(-3) \\
 &= 6 - 6 = 0
 \end{aligned}$$

Conclusion

(iv) Let

$$A = \begin{vmatrix} 2 & -1 & -2 \\ 0 & 2 & -1 \\ 3 & -5 & 0 \end{vmatrix}$$

Hence,

$$\begin{aligned}
 |A| &= 2 \begin{vmatrix} 2 & -1 \\ -5 & 0 \end{vmatrix} - 0 \begin{vmatrix} -1 & -2 \\ -5 & 0 \end{vmatrix} + 3 \begin{vmatrix} -1 & -2 \\ 2 & -1 \end{vmatrix} \\
 &= 2(0 - 5) - 0 + 3(1 + 4) \\
 &= -10 + 15 = 5
 \end{aligned}$$

Q

6

Verifying
 $|A| = |A^T|$



If $A = \begin{bmatrix} 1 & 1 & -2 \\ 2 & 1 & -3 \\ 5 & 4 & -9 \end{bmatrix}$, find $|A|$

Given



Let

$$A = \begin{bmatrix} 1 & 1 & -2 \\ 2 & 1 & -3 \\ 5 & 4 & -9 \end{bmatrix}$$

Substitute

Hence,

$$\begin{aligned}
 |A| &= 1 \begin{vmatrix} 1 & -3 \\ 4 & -9 \end{vmatrix} - 1 \begin{vmatrix} 2 & -3 \\ 5 & -9 \end{vmatrix} - 2 \begin{vmatrix} 2 & 1 \\ 5 & 4 \end{vmatrix} \\
 &= 1(-9 + 12) - 1(-18 + 15) - 2(8 - 5) \\
 &= 1(3) - 1(-3) - 2(3) \\
 &= 3 + 3 - 6 \\
 &= 0
 \end{aligned}$$

Q

7

Solving for x
 from equal
 2x2
 determinants



Find values of x, if

Part (i)

$$\begin{vmatrix} 2 & 4 \\ 5 & 1 \end{vmatrix} = \begin{vmatrix} 2x & 4 \\ 6 & x \end{vmatrix}$$

Part (ii)

$$\begin{vmatrix} 2 & 3 \\ 4 & 5 \end{vmatrix} = \begin{vmatrix} x & 3 \\ 2x & 5 \end{vmatrix}$$

Given



$$(i) \begin{vmatrix} 2 & 4 \\ 5 & 1 \end{vmatrix} = \begin{vmatrix} 2x & 4 \\ 6 & x \end{vmatrix}$$

Therefore,

$$\begin{aligned}\Rightarrow 2 \times 1 - 5 \times 4 &= 2x \times x - 6 \times 4 \\ \Rightarrow 2 - 20 &= 2x^2 - 24 \\ \Rightarrow 2x^2 &= 6 \\ \Rightarrow x^2 &= 3 \\ \Rightarrow x &= \pm \sqrt{3}\end{aligned}$$

Conclusion

$$(ii) \begin{vmatrix} 2 & 3 \\ 4 & 5 \end{vmatrix} = \begin{vmatrix} x & 3 \\ 2x & 5 \end{vmatrix}$$

Therefore,

$$\begin{aligned}\Rightarrow 2 \times 5 - 3 \times 4 &= x \times 5 - 3 \times 2x \\ \Rightarrow 10 - 12 &= 5x - 6x \\ \Rightarrow -2 &= -x \\ \Rightarrow x &= 2\end{aligned}$$

Q

8

MCQ - solving for x from equal 2x2 determinants

(())

If $\begin{vmatrix} x & 2 \\ 18 & x \end{vmatrix} = \begin{vmatrix} 6 & 2 \\ 18 & 6 \end{vmatrix}$, then x is equal to (A) 6 (B) ± 6 (C) -6 (D) 0

Substitute

$$\begin{vmatrix} x & 2 \\ 18 & x \end{vmatrix} = \begin{vmatrix} 6 & 2 \\ 18 & 6 \end{vmatrix}$$

Therefore,

$$\begin{aligned}\Rightarrow x^2 - 36 &= 36 - 36 \\ \Rightarrow x^2 - 36 &= 0 \\ \Rightarrow x^2 &= 36 \\ \Rightarrow x &= \pm 6\end{aligned}$$

Conclusion

Thus, the correct option is B.

Q

9

Area of a triangle from its vertices

()

Find area of the triangle with vertices at the point given in each of the following:

Part (i)

(1, 0), (6, 0), (4, 3)

Part (ii)

(2, 7), (1, 1), (10, 8)

Part (iii)

(-2, -3), (3, 2), (-1, -8)

Given

(i) The area of the triangle with vertices (1,0),(6,0),(4,3) is given by the relation,

$$\begin{aligned}\Delta &= \frac{1}{2} \begin{vmatrix} 1 & 0 & 1 \\ 6 & 0 & 1 \\ 4 & 3 & 1 \end{vmatrix} \\ &= \frac{1}{2} [1(0-3) - 0(6-4) + 1(18-0)] \\ &= \frac{1}{2} [-3 + 18] \\ &= \frac{1}{2} [15] \\ &= \frac{15}{2}\end{aligned}$$

Hence, area of the triangle is $\frac{15}{2}$ square units.

Substitute

(ii) The area of the triangle with vertices (2,7),(1,1),(10,8) is given by the relation,

$$\begin{aligned}\Delta &= \frac{1}{2} \begin{vmatrix} 2 & 7 & 1 \\ 1 & 1 & 1 \\ 10 & 8 & 1 \end{vmatrix} \\ &= \frac{1}{2} [2(1-8)-7(1-10)+1(8-10)] \\ &= \frac{1}{2} [2(-7)-7(-9)+1(-2)] \\ &= \frac{1}{2} [-14+63-2] \\ &= \frac{1}{2} [47] \\ &= \frac{47}{2}\end{aligned}$$

Hence, area of the triangle is $\frac{47}{2}$ square units.

Conclusion

(iii) The area of the triangle with vertices (-2,-3),(3,2),(-1,-8) is given by the relation,

$$\begin{aligned}\Delta &= \frac{1}{2} \begin{vmatrix} -2 & -3 & 1 \\ 3 & 2 & 1 \\ -1 & -8 & 1 \end{vmatrix} \\ &= \frac{1}{2} [-2(2+8)+3(3+1)+1(-24+2)] \\ &= \frac{1}{2} [-2(10)+3(4)+1(-22)] \\ &= \frac{1}{2} [-20+12-22] \\ &= -\frac{1}{2} [30] \\ &= -15\end{aligned}$$

Hence, area of the triangle is 15 square units.

Q**10**

Showing
three points
are collinear

**Show that points**

A(a, b+c), B(b, c+a), C(c, a+b) are collinear.

Substitute

The area of the triangle with vertices A(a, b+c), B(b, c+a), C(c, a+b) is given by the absolute value of the relation:

$$\begin{aligned}\Delta &= \frac{1}{2} \begin{vmatrix} a & b+c & 1 \\ b & c+a & 1 \\ c & a+b & 1 \end{vmatrix} \\ &= \frac{1}{2} \begin{vmatrix} a & b+c & 1 \\ b-a & a-b & 0 \\ c-a & a-c & 0 \end{vmatrix} [R_2 \rightarrow R_2-R_1 \text{ and } R_3 \rightarrow R_3-R_1] \\ &= \frac{1}{2} (a-b)(c-a) \begin{vmatrix} a & b+c & 1 \\ -1 & 1 & 0 \\ 1 & -1 & 0 \end{vmatrix} \\ &= \frac{1}{2} (a-b)(c-a) \begin{vmatrix} a & b+c & 1 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \end{vmatrix} [R_3 \rightarrow R_3+R_2]\end{aligned}$$

Conclusion

Thus, the area of the triangle formed by points is zero.

Hence, the points are collinear.

Q**11**

Finding k
from a given
triangle area



Find values of k if area of triangle is 4 sq. units and vertices are

Part (i)

(k, 0),(4,0),(0,2)

Part (ii)

(-2, 0), (0, 4), (0, k)

Key formula

We know that the area of a triangle with vertices $(x_1, y_1), (x_2, y_2)$ and (x_3, y_3) is the absolute value of the determinant (Δ).

Here,

$$\Delta = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

Given

It is given that the area of triangle is 4 square units.

Hence, $\Delta = \pm 4$

Substitute

(i) The area of the triangle with vertices (k, 0),(4,0),(0,2) is given by the relation,

$$\begin{aligned} \Delta &= \frac{1}{2} \begin{vmatrix} k & 0 & 1 \\ 4 & 0 & 1 \\ 0 & 2 & 1 \end{vmatrix} \\ &= \frac{1}{2} [k(0-2)-0(4-0)+1(8-0)] \\ &= \frac{1}{2} [-2k+8] \\ &= -k+4 \end{aligned}$$

Therefore, $-k+4 = \pm 4$

When $-k+4 = -4$

Then $k = 8$

When $-k+4 = 4$

Then $k = 0$

Hence, $k = 0, 8$

Conclusion

(ii) The area of the triangle with vertices (-2,0),(0,4),(0, k) is given by the relation,

$$\begin{aligned} \Delta &= \frac{1}{2} \begin{vmatrix} -2 & 0 & 1 \\ 0 & 4 & 1 \\ 0 & k & 1 \end{vmatrix} \\ &= \frac{1}{2} [-2(4-k)] \\ &= k-4 \end{aligned}$$

Therefore, $-k+4 = \pm 4$

When $k-4 = 4$

Then $k = 8$

When $k-4 = -4$

Then $k = 0$

Hence, $k = 0, 8$

Q**12**

Equation of a line via determinants

(()

Part (i)

Find equation of line joining (1,2) and (3,6) using determinants.

Part (ii)

Find equation of line joining (3,1) and (9,3) using determinants.**Given**

(i) Let P(x, y) be any point on the line joining points A(1,2) and B(3,6).

Then, the points A, B and P are collinear.

Hence, the area of triangle A B P will be zero.

Therefore,

$$\Rightarrow \frac{1}{2} \begin{vmatrix} 1 & 2 & 1 \\ 3 & 6 & 1 \\ x & y & 1 \end{vmatrix} = 0$$

$$\Rightarrow \frac{1}{2} [1(6-y)-2(3-x)+1(3y-6x)] = 0$$

$$\Rightarrow 6-y-6+2x+3y-6x = 0$$

$$\Rightarrow 2y-4x = 0$$

$$\Rightarrow y = 2x$$

Thus, the equation of the line joining the given points is $y = 2x$.**Conclusion**

(ii) Let P(x, y) be any point on the line joining points A(3,1) and B(9,3).

Then, the points A, B and P are collinear.

Hence, the area of triangle A B P will be zero.

Therefore,

$$\Rightarrow \frac{1}{2} \begin{vmatrix} 3 & 1 & 1 \\ 9 & 3 & 1 \\ x & y & 1 \end{vmatrix} = 0$$

$$\Rightarrow \frac{1}{2} [3(3-y)-1(9-x)+1(9y-3x)] = 0$$

$$\Rightarrow 9-3y-9+x+9y-3x = 0$$

$$\Rightarrow 6y-2x = 0$$

$$\Rightarrow x-3y = 0$$

Thus, the equation of the line joining the given points is $x-3y = 0$.**Q****13**

MCQ - finding k from a given triangle area

✧

If area of triangle is 35 sq units with vertices (2,-6),(5,4) and (k, 4). Then k is (A) 12 (B) -2 (C) -12, -2 (D) 12,-2**Substitute**

The area of the triangle with vertices (2,-6),(5,4),(k, 4) is given by the relation,

$$\Delta = \frac{1}{2} \begin{vmatrix} 2 & -6 & 1 \\ 5 & 4 & 1 \\ k & 4 & 1 \end{vmatrix}$$

$$= \frac{1}{2} [2(4-4)+6(5-k)+1(20-4k)]$$

$$= \frac{1}{2} [30-6k+20-4k]$$

$$= \frac{1}{2} [50-10k]$$

$$= 25-5k$$

It is given that the area of the triangle is 35 square units.

Hence, $\Delta = \pm 35$.

Therefore,

$$\Rightarrow 25-5k = \pm 35$$

$$\Rightarrow 5(5-k) = \pm 35$$

$$\Rightarrow 5-k = \pm 7$$

Conclusion

When, $5-k = -7$

Then, $k = 12$

When, $5-k = 7$

Then, $k = -2$

Hence, $k = 12, -2$

Thus, the correct option is D.

Q

14

Minors and
cofactors of a
2x2
determinant



Write Minors and Cofactors of the elements of following determinants:

Part (i)

$$\begin{vmatrix} 2 & -4 \\ 0 & 3 \end{vmatrix}$$

Part (ii)

$$\begin{vmatrix} a & c \\ b & d \end{vmatrix}$$

Given



(i) The given determinant is $\begin{vmatrix} 2 & -4 \\ 0 & 3 \end{vmatrix}$

Minor of element a_{ij} is M_{ij} .

$$M_{11} = \text{minor of element } a_{11} = 3$$

$$M_{12} = \text{minor of element } a_{12} = 0$$

$$M_{21} = \text{minor of element } a_{21} = -4$$

$$M_{22} = \text{minor of element } a_{22} = 2$$

Cofactor of a_{ij} is $A_{ij} = (-1)^{i+j} M_{ij}$

$$A_{11} = (-1)^{1+1} M_{11} = (-1)^2(3) = 3$$

$$A_{12} = (-1)^{1+2} M_{12} = (-1)^3(0) = 0$$

$$A_{21} = (-1)^{2+1} M_{21} = (-1)^3(-4) = 4$$

$$A_{22} = (-1)^{2+2} M_{22} = (-1)^4(2) = 2$$

Conclusion

(ii) The given determinant is $\begin{vmatrix} a & c \\ b & d \end{vmatrix}$

Minor of element a_{ij} is M_{ij} .

$$M_{11} = \text{minor of element } a_{11} = d$$

$$M_{12} = \text{minor of element } a_{12} = b$$

$$M_{21} = \text{minor of element } a_{21} = c$$

$$M_{22} = \text{minor of element } a_{22} = a$$

Cofactor of a_{ij} is $A_{ij} = (-1)^{i+j} M_{ij}$

$$A_{11} = (-1)^{1+1} M_{11} = (-1)^2(d) = d$$

$$A_{12} = (-1)^{1+2} M_{12} = (-1)^3(b) = -b$$

$$A_{21} = (-1)^{2+1} M_{21} = (-1)^3(c) = -c$$

$$A_{22} = (-1)^{2+2} M_{22} = (-1)^4(a) = a$$

Q

15

Minors and
cofactors of a
3x3
determinant



Write Minors and Cofactors of the elements of following determinants:

Part (i)

$$\begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

Part (ii)

$$\begin{vmatrix} 1 & 0 & 4 \\ 3 & 5 & -1 \\ 0 & 1 & 2 \end{vmatrix}$$

Given



(i) The given determinant is $\begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$

Minor of element a_{ij} is M_{ij} .

$$M_{11} = \text{minor of element } a_{11} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1$$

$$M_{12} = \text{minor of element } a_{12} = \begin{vmatrix} 0 & 0 \\ 0 & 1 \end{vmatrix} = 0$$

$$M_{13} = \text{minor of element } a_{13} = \begin{vmatrix} 0 & 1 \\ 0 & 0 \end{vmatrix} = 0$$

$$M_{21} = \text{minor of element } a_{21} = \begin{vmatrix} 0 & 0 \\ 0 & 1 \end{vmatrix} = 0$$

$$M_{22} = \text{minor of element } a_{22} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1$$

$$M_{23} = \text{minor of element } a_{23} = \begin{vmatrix} 1 & 0 \\ 0 & 0 \end{vmatrix} = 0$$

$$M_{31} = \text{minor of element } a_{31} = \begin{vmatrix} 0 & 0 \\ 1 & 0 \end{vmatrix} = 0$$

$$M_{32} = \text{minor of element } a_{32} = \begin{vmatrix} 1 & 0 \\ 0 & 0 \end{vmatrix} = 0$$

$$M_{33} = \text{minor of element } a_{33} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1$$

Cofactor of a_{ij} is $A_{ij} = (-1)^{i+j} M_{ij}$

$$A_{11} = (-1)^{1+1} M_{11} = (-1)^2(1) = 1$$

$$A_{12} = (-1)^{1+2} M_{12} = (-1)^3(0) = 0$$

$$A_{13} = (-1)^{1+3} M_{13} = (-1)^4(0) = 0$$

$$A_{21} = (-1)^{2+1} M_{21} = (-1)^3(0) = 0$$

$$A_{22} = (-1)^{2+2} M_{22} = (-1)^4(1) = 1$$

$$A_{23} = (-1)^{2+3} M_{23} = (-1)^5(0) = 0$$

$$A_{31} = (-1)^{3+1} M_{31} = (-1)^4(0) = 0$$

$$A_{32} = (-1)^{3+2} M_{32} = (-1)^5(0) = 0$$

$$A_{33} = (-1)^{3+3} M_{33} = (-1)^6(1) = 1$$

Conclusion

(ii) The given determinant is $\begin{vmatrix} 1 & 0 & 4 \\ 3 & 5 & -1 \\ 0 & 1 & 2 \end{vmatrix}$

Minor of element a_{ij} is M_{ij} .

$$M_{11} = \text{minor of element } a_{11} = \begin{vmatrix} 5 & -1 \\ 1 & 2 \end{vmatrix} = 11$$

$$M_{12} = \text{minor of element } a_{12} = \begin{vmatrix} 3 & -1 \\ 0 & 2 \end{vmatrix} = 6$$

$$M_{13} = \text{minor of element } a_{13} = \begin{vmatrix} 3 & 5 \\ 0 & 1 \end{vmatrix} = 3$$

$$M_{21} = \text{minor of element } a_{21} = \begin{vmatrix} 0 & 4 \\ 1 & 2 \end{vmatrix} = -4$$

$$M_{22} = \text{minor of element } a_{22} = \begin{vmatrix} 1 & 4 \\ 0 & 2 \end{vmatrix} = 2$$

$$M_{23} = \text{minor of element } a_{23} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1$$

$$M_{31} = \text{minor of element } a_{31} = \begin{vmatrix} 0 & 4 \\ 5 & -1 \end{vmatrix} = -20$$

$$M_{32} = \text{minor of element } a_{32} = \begin{vmatrix} 1 & 4 \\ 3 & -1 \end{vmatrix} = -13$$

$$M_{33} = \text{minor of element } a_{33} = \begin{vmatrix} 1 & 0 \\ 3 & 5 \end{vmatrix} = 5$$

Cofactor of a_{ij} is $A_{ij} = (-1)^{i+j} M_{ij}$

$$A_{11} = (-1)^{1+1} M_{11} = (-1)^2(11) = 11$$

$$A_{12} = (-1)^{1+2} M_{12} = (-1)^3(6) = -6$$

$$A_{13} = (-1)^{1+3} M_{13} = (-1)^4(3) = 3$$

$$A_{21} = (-1)^{2+1} M_{21} = (-1)^3(-4) = 4$$

$$A_{22} = (-1)^{2+2} M_{22} = (-1)^4(2) = 2$$

$$A_{23} = (-1)^{2+3} M_{23} = (-1)^5(1) = -1$$

$$A_{31} = (-1)^{3+1} M_{31} = (-1)^4(-20) = -20$$

$$A_{32} = (-1)^{3+2} M_{32} = (-1)^5(-13) = 13$$

$$A_{33} = (-1)^{3+3} M_{33} = (-1)^6(5) = 5$$

Q

16

Evaluating a determinant via row cofactors

(())

Using Cofactors of elements of second row, evaluate $\Delta =$

$$\begin{vmatrix} 5 & 3 & 8 \\ 2 & 0 & 1 \\ 1 & 2 & 3 \end{vmatrix}$$

Given

The given determinant is $\begin{vmatrix} 5 & 3 & 8 \\ 2 & 0 & 1 \\ 1 & 2 & 3 \end{vmatrix}$

Substitute

$$M_{21} = \text{minor of element } a_{21} = \begin{vmatrix} 3 & 8 \\ 2 & 3 \end{vmatrix} = -7$$

$$A_{21} = (-1)^{2+1} M_{21} = (-1)^3(-7) = 7$$

$$M_{22} = \text{minor of element } a_{22} = \begin{vmatrix} 5 & 8 \\ 1 & 3 \end{vmatrix} = 15-8 = 7$$

$$A_{22} = (-1)^{2+2} M_{22} = (-1)^4(7) = 7$$

$$M_{23} = \text{minor of element } a_{23} = \begin{vmatrix} 5 & 3 \\ 1 & 2 \end{vmatrix} = 7$$

$$A_{23} = (-1)^{2+3} M_{21} = (-1)^5(7) = -7$$

Key formula

We know that Δ equals the sum of the products of the second row's elements and their corresponding cofactors.

Conclusion

Therefore,

$$\begin{aligned}\Delta &= a_{21} A_{21} + a_{22} A_{22} + a_{23} A_{23} \\ &= 2(7) + 0(7) + 1(-7) \\ &= 14 - 7 \\ &= 7\end{aligned}$$

Q

17

Evaluating a
determinant
via column
cofactors



Using Cofactors of elements of third column, evaluate $\Delta = \begin{vmatrix} 1 & x & y & z \\ 1 & y & z & x \\ 1 & z & x & y \end{vmatrix}$.

Given

The given determinant is $\begin{vmatrix} 1 & x & y & z \\ 1 & y & z & x \\ 1 & z & x & y \end{vmatrix}$

Substitute

Therefore,

$$M_{13} = \begin{vmatrix} 1 & y \\ 1 & z \end{vmatrix} = z - y$$

$$M_{23} = \begin{vmatrix} 1 & x \\ 1 & z \end{vmatrix} = z - x$$

$$M_{33} = \begin{vmatrix} 1 & x \\ 1 & y \end{vmatrix} = y - x$$

$$A_{13} = (-1)^{1+3} M_{13} = (-1)^4 (z - y) = z - y$$

$$A_{23} = (-1)^{2+3} M_{23} = (-1)^5 (z - x) = -(z - x) = x - z$$

$$A_{33} = (-1)^{3+3} M_{33} = (-1)^6 (y - x) = y - x$$

Key formula

We know that Δ equals the sum of the products of the third column's elements and their corresponding cofactors.

Conclusion

Therefore,

$$\begin{aligned}\Delta &= a_{13} A_{13} + a_{23} A_{23} + a_{33} A_{33} \\ &= y z(z - y) + z x(x - z) + x y(y - x) \\ &= y z^2 - y^2 z + x^2 z - x z^2 + x y^2 - x^2 y \\ &= (x^2 z - y^2 z) + (y z^2 - x z^2) + (x y^2 - x^2 y) \\ &= z(x^2 - y^2) + z^2(y - x) + x y(y - x) \\ &= z(x - y)(x + y) + z^2(y - x) + x y(y - x) \\ &= (x - y)[z x + z y - z^2 - x y] \\ &= (x - y)[z(x - z) + y(z - x)] \\ &= (x - y)(z - x)[-z + y] \\ &= (x - y)(y - z)(z - x)\end{aligned}$$

Hence,

$$\Delta = (x - y)(y - z)(z - x)$$

Q

18

Proving a determinant equals its cofactor-matrix determinant squared



If $\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$ and A_{ij} is Cofactors of a_{ij} , then value of Δ is given by (A)

$a_{11} A_{31} + a_{12} A_{32} + a_{13} A_{33}$ (B) $a_{11} A_{11} + a_{12} A_{21} + a_{13} A_{31}$ (C)

$a_{21} A_{11} + a_{22} A_{12} + a_{23} A_{13}$ (D) $a_{11} A_{11} + a_{21} A_{21} + a_{31} A_{31}$

4.5 Adjoint and Inverse of a Matrix In the previous chapter, we have studied inverse of a matrix. In this section, we shall discuss the condition for existence of inverse of a matrix. To

find inverse of a matrix A , i.e., A^{-1} we shall first define adjoint of a matrix.

4.5.1 Adjoint of a matrix

Definition 3 The adjoint of a square matrix $A = [a_{ij}]_{n \times n}$ is defined as the transpose of the matrix $[A_{ij}]_{n \times n}$, where A_{ij} is the cofactor of the element a_{ij} .

Adjoint of the matrix A is denoted by $\text{adj } A$.

Let

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Then

$$\text{adj } A = \text{Transpose of } \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix} = \begin{bmatrix} A_{11} & A_{21} & A_{31} \\ A_{12} & A_{22} & A_{32} \\ A_{13} & A_{23} & A_{33} \end{bmatrix}$$

Key formula

We know that Δ equals the sum of the products of the elements of a column or row and their corresponding cofactors.



Conclusion

$$\Delta = a_{11} A_{11} + a_{21} A_{21} + a_{31} A_{31}$$

Thus, the correct option is D.

Q

19

Finding the adjoint of a 2x2 matrix



Find adjoint of each of the matrices in Exercises 1 and 2.

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

Given

Let

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$



Substitute

Then,

$$A_{11} = 4 \quad A_{12} = -3$$

$$A_{21} = -2 \quad A_{22} = 1$$

Conclusion

Therefore,

$$\begin{aligned} \text{adj } A &= \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} \\ &= \begin{pmatrix} 4 & -2 \\ -3 & 1 \end{pmatrix} \end{aligned}$$

Q**20**

Finding the
adjoint of a
3x3 matrix

(())

Find adjoint of each of the matrices in Exercises 1 and 2.

$$\begin{bmatrix} 1 & -1 & 2 \\ 2 & 3 & 5 \\ -2 & 0 & 1 \end{bmatrix}$$

Given

Let

$$A = \begin{pmatrix} 1 & -1 & 2 \\ 2 & 3 & 5 \\ -2 & 0 & 1 \end{pmatrix}$$

Substitute

Then,

$$\begin{aligned} A_{11} &= \begin{vmatrix} 3 & 5 \\ 0 & 1 \end{vmatrix} = 3 & A_{12} &= - \begin{vmatrix} 2 & 5 \\ -2 & 1 \end{vmatrix} = -12 & A_{13} &= \begin{vmatrix} 2 & 3 \\ -2 & 0 \end{vmatrix} = 6 \\ A_{21} &= - \begin{vmatrix} -1 & 2 \\ 0 & 1 \end{vmatrix} = -1 & A_{22} &= \begin{vmatrix} 1 & 2 \\ -2 & 1 \end{vmatrix} = 5 & A_{23} &= - \begin{vmatrix} 1 & -1 \\ -2 & 0 \end{vmatrix} = 2 \\ A_{31} &= \begin{vmatrix} -1 & 2 \\ 3 & 5 \end{vmatrix} = -11 & A_{32} &= - \begin{vmatrix} 1 & 2 \\ 2 & 5 \end{vmatrix} = -1 & A_{33} &= \begin{vmatrix} 1 & -1 \\ 2 & 3 \end{vmatrix} = 5 \end{aligned}$$

Conclusion

Therefore,

$$\text{adj } A = \begin{pmatrix} 3 & 1 & -11 \\ -12 & 5 & -1 \\ 6 & 2 & 5 \end{pmatrix}$$

Q**21**

Verifying
 $A(\text{adj } A) =$
 $(\text{adj } A)A =$
 $|A| I$ (2x2)



Verify $A (\text{adj } A) = (\text{adj } A) A = |A| I$ in Exercises 3 and 4

$$\begin{bmatrix} 2 & 3 \\ -4 & -6 \end{bmatrix}$$

Given

Let

$$A = \begin{pmatrix} 2 & 3 \\ -4 & -6 \end{pmatrix}$$

Substitute

Then,

$$\begin{aligned} |A| &= -12 - (-12) \\ &= 0 \end{aligned}$$

Also,

$$\begin{aligned} |A| I &= 0 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \end{aligned}$$

Now,

$$\begin{aligned} A_{11} &= -6 & A_{12} &= 4 \\ A_{21} &= -3 & A_{22} &= 2 \end{aligned}$$

Hence,

$$\text{adj } A = \begin{pmatrix} -6 & -3 \\ 4 & 2 \end{pmatrix}$$

Now,

$$\begin{aligned}
 A(\text{adj } A) &= \begin{pmatrix} 2 & 3 \\ -4 & -6 \end{pmatrix} \begin{pmatrix} -6 & -3 \\ 4 & 2 \end{pmatrix} \\
 &= \begin{pmatrix} -12+12 & -6+6 \\ 24-24 & 12-12 \end{pmatrix} \\
 &= \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}
 \end{aligned}$$

Also,

$$\begin{aligned}
 (\text{adj } A) A &= \begin{pmatrix} -6 & -3 \\ 4 & 2 \end{pmatrix} \begin{pmatrix} 2 & 3 \\ -4 & -6 \end{pmatrix} \\
 &= \begin{pmatrix} -12+12 & -18+18 \\ 8-8 & 12-12 \end{pmatrix} \\
 &= \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}
 \end{aligned}$$

Conclusion

Hence, $A(\text{adj } A) = (\text{adj } A) A = |A| I$.

Q

22

Verifying
 $A(\text{adj } A) =$
 $(\text{adj } A)A =$
 $|A|I$ (3x3)



Verify $A(\text{adj } A) = (\text{adj } A)A = |A| I$ in Exercises 3 and 4

$$\begin{bmatrix} 1 & -1 & 2 \\ 3 & 0 & -2 \\ 1 & 0 & 3 \end{bmatrix}$$

Given



$$A = \begin{pmatrix} 1 & -1 & 2 \\ 3 & 0 & -2 \\ 1 & 0 & 3 \end{pmatrix}$$

Substitute

Then,

$$\begin{aligned}
 |A| &= 1(0-0)+1(9+2)+2(0-0) \\
 &= 11
 \end{aligned}$$

Also,

$$\begin{aligned}
 |A| I &= 11 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\
 &= \begin{pmatrix} 11 & 0 & 0 \\ 0 & 11 & 0 \\ 0 & 0 & 11 \end{pmatrix}
 \end{aligned}$$

Now,

$$\begin{aligned}
 A_{11} &= 0 \quad A_{12} = -11 \quad A_{13} = 0 \\
 A_{21} &= 3 \quad A_{22} = 1 \quad A_{23} = -1 \\
 A_{31} &= 2 \quad A_{32} = 8 \quad A_{33} = 3
 \end{aligned}$$

Hence,

$$\text{adj } A = \begin{pmatrix} 0 & 3 & 2 \\ -11 & 1 & 8 \\ 0 & -1 & 3 \end{pmatrix}$$

Now,

$$\begin{aligned}
 A(\text{adj } A) &= \begin{pmatrix} 1 & -1 & 2 \\ 3 & 0 & -2 \\ 1 & 0 & 3 \end{pmatrix} \begin{pmatrix} 0 & 3 & 2 \\ -11 & 1 & 8 \\ 0 & -1 & 3 \end{pmatrix} \\
 &= \begin{pmatrix} 0+11+0 & 3-1-2 & 2-8+6 \\ 0+0+0 & 9+0+2 & 6+0-6 \\ 0+0+0 & 3+0-3 & 2+0+9 \end{pmatrix} = \begin{pmatrix} 11 & 0 & 0 \\ 0 & 11 & 0 \\ 0 & 0 & 11 \end{pmatrix}
 \end{aligned}$$

Also,

$$\begin{aligned}
 (\text{adj } A) A &= \begin{pmatrix} 0 & 3 & 2 \\ -11 & 1 & 8 \\ 0 & -1 & 3 \end{pmatrix} \begin{pmatrix} 1 & -1 & 2 \\ 3 & 0 & -2 \\ 1 & 0 & 3 \end{pmatrix} \\
 &= \begin{pmatrix} 0+9+2 & 0+0+0 & 0-6+6 \\ -11+3+8 & 11+0+0 & -22-2+24 \\ 0-3+3 & 0+0+0 & 2+0+9 \end{pmatrix} \\
 &= \begin{pmatrix} 11 & 0 & 0 \\ 0 & 11 & 0 \\ 0 & 0 & 11 \end{pmatrix}
 \end{aligned}$$

Conclusion

Hence, $A(\text{adj } A) = (\text{adj } A) A = |A| I$.

Q

23

Finding the
inverse of a
2x2 matrix



Find the inverse of each of the matrices (if it exists) given in Exercises 5 to 11.

$$\begin{bmatrix} 2 & -2 \\ 4 & 3 \end{bmatrix}$$

Given

$$\text{Let } A = \begin{pmatrix} 2 & -2 \\ 4 & 3 \end{pmatrix}$$

Substitute

Then,

$$\begin{aligned}
 |A| &= 6+8 \\
 &= 14
 \end{aligned}$$

Now,

$$A_{11} = 3 \quad A_{12} = -4$$

$$A_{21} = 2 \quad A_{22} = 2$$

Therefore,

$$\text{adj } A = \begin{pmatrix} 3 & 2 \\ -4 & 2 \end{pmatrix}$$

Conclusion

Hence,

$$\begin{aligned}
 A^{-1} &= \frac{1}{|A|} \text{adj } A \\
 &= \frac{1}{14} \begin{pmatrix} 3 & 2 \\ -4 & 2 \end{pmatrix}
 \end{aligned}$$

Q**24**

Finding the
inverse of a
2x2 matrix

(())

Find the inverse of each of the matrices (if it exists) given in Exercises 5 to 11.

$$\begin{bmatrix} -1 & 5 \\ -3 & 2 \end{bmatrix}$$

Given

$$\text{Let } A = \begin{pmatrix} -1 & 5 \\ -3 & 2 \end{pmatrix}$$

Substitute

Then,

$$|A| = -2 + 15 = 13$$

Now,

$$A_{11} = 2 \quad A_{12} = 3$$

$$A_{21} = -5 \quad A_{22} = -1$$

Therefore,

$$\text{adj } A = \begin{pmatrix} 2 & -5 \\ 3 & -1 \end{pmatrix}$$

Conclusion

Hence,

$$\begin{aligned} A^{-1} &= \frac{1}{|A|} \text{adj } A \\ &= \frac{1}{13} \begin{pmatrix} 2 & -5 \\ 3 & -1 \end{pmatrix} \end{aligned}$$

Q**25**

Finding the
inverse of a
3x3 matrix
(upper
triangular)

(*)

Find the inverse of each of the matrices (if it exists) given in Exercises 5 to 11.

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 4 \\ 0 & 0 & 5 \end{bmatrix}$$

Given

Let

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 2 & 4 \\ 0 & 0 & 5 \end{pmatrix}$$

Substitute

Then,

$$\begin{aligned} |A| &= 1(10-0) - 2(0-0) + 3(0-0) \\ &= 10 \end{aligned}$$

Now,

$$A_{11} = 10 \quad A_{12} = 0 \quad A_{13} = 0$$

$$A_{21} = -10 \quad A_{22} = 5 \quad A_{23} = 0$$

$$A_{31} = 2 \quad A_{32} = -4 \quad A_{33} = 2$$

Therefore,

$$\text{adj } A = \begin{pmatrix} 10 & -10 & 2 \\ 0 & 5 & -4 \\ 0 & 0 & 2 \end{pmatrix}$$

Hence,

Conclusion

$$A^{-1} = \frac{1}{|A|} \text{adj } A$$

$$= \frac{1}{10} \begin{pmatrix} 10 & -10 & 2 \\ 0 & 5 & -4 \\ 0 & 0 & 2 \end{pmatrix}$$

Q**26**

Finding the
inverse of a
3x3 matrix



Find the inverse of each of the matrices (if it exists) given in Exercises 5 to 11.

$$\begin{bmatrix} 1 & 0 & 0 \\ 3 & 3 & 0 \\ 5 & 2 & -1 \end{bmatrix}$$

Given

Let

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 3 & 0 \\ 5 & 2 & -1 \end{pmatrix}$$

Substitute

Then,

$$|A| = 1(-3-0)-0+0 = -3$$

Now,

$$A_{11} = -3 \quad A_{12} = 3 \quad A_{13} = -9$$

$$A_{21} = 0 \quad A_{22} = -1 \quad A_{23} = -2$$

$$A_{31} = 0 \quad A_{32} = 0 \quad A_{33} = 3$$

Therefore,

$$\text{adj } A = \begin{pmatrix} -3 & 0 & 0 \\ 3 & -1 & 0 \\ -9 & -2 & 3 \end{pmatrix}$$

Hence,

Conclusion

$$A^{-1} = \frac{1}{|A|} \text{adj } A$$

$$= \frac{-1}{3} \begin{pmatrix} -3 & 0 & 0 \\ 3 & -1 & 0 \\ -9 & -2 & 3 \end{pmatrix}$$

Q**27**

Finding the
inverse of a
3x3 matrix



Find the inverse of each of the matrices (if it exists) given in Exercises 5 to 11.

$$\begin{bmatrix} 2 & 1 & 3 \\ 4 & -1 & 0 \\ -7 & 2 & 1 \end{bmatrix}$$

Given

Let

$$A = \begin{pmatrix} 2 & 1 & 3 \\ 4 & -1 & 0 \\ -7 & 2 & 1 \end{pmatrix}$$

Substitute

Then,

$$|A| = 2(-1-0)-1(4-0)+3(8-7)$$

$$= 2(-1)-1(4)+3(1)$$

$$= -3$$

Now,

$$\begin{aligned} A_{11} &= -1 \quad A_{12} = -4 \quad A_{13} = 1 \\ A_{21} &= 5 \quad A_{22} = 23 \quad A_{23} = -11 \\ A_{31} &= 3 \quad A_{32} = 12 \quad A_{33} = -6 \end{aligned}$$

Therefore,

$$\text{adj } A = \begin{bmatrix} -1 & 5 & 3 \\ -4 & 23 & 12 \\ 1 & -11 & -6 \end{bmatrix}$$

Hence,

Conclusion

$$\begin{aligned} A^{-1} &= \frac{1}{|A|} \text{adj } A \\ &= -\frac{1}{3} \begin{bmatrix} -1 & 5 & 3 \\ -4 & 23 & 12 \\ 1 & -11 & -6 \end{bmatrix} \end{aligned}$$

Q

28

Finding the
inverse of a
3x3 matrix

()

Find the inverse of each of the matrices (if it exists) given in Exercises 5 to 11.

$$\begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & -3 \\ 3 & -2 & 4 \end{bmatrix}$$

Given



Let

$$A = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & -3 \\ 3 & -2 & 4 \end{bmatrix}$$

Substitute

Then, expanding along C_1 ,

$$\begin{aligned} |A| &= 1(8-6)-0+3(3-4) = 2-3 \\ &= -1 \end{aligned}$$

Now,

$$\begin{aligned} A_{11} &= 2 \quad A_{12} = -9 \quad A_{13} = -6 \\ A_{21} &= 0 \quad A_{22} = -2 \quad A_{23} = -1 \\ A_{31} &= -1 \quad A_{32} = 3 \quad A_{33} = 2 \end{aligned}$$

Therefore,

$$\text{adj } A = \begin{bmatrix} 2 & 0 & -1 \\ -9 & -2 & 3 \\ -6 & -1 & 2 \end{bmatrix}$$

Hence,

Conclusion

$$\begin{aligned} A^{-1} &= \frac{1}{|A|} \text{adj } A \\ &= -1 \begin{bmatrix} 2 & 0 & -1 \\ -9 & -2 & 3 \\ -6 & -1 & 2 \end{bmatrix} \\ &= \begin{bmatrix} -2 & 0 & 1 \\ 9 & 2 & -3 \\ 6 & 1 & -2 \end{bmatrix} \end{aligned}$$

Q

29

Finding the
inverse of a
3x3 rotation-
type matrix



Find the inverse of each of the matrices (if it exists) given in Exercises 5 to

11.
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \alpha & \sin \alpha \\ 0 & \sin \alpha & -\cos \alpha \end{bmatrix}$$

Given



Let

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \alpha & \sin \alpha \\ 0 & \sin \alpha & -\cos \alpha \end{bmatrix}$$

Substitute

Then,

$$|A| = 1(-\cos^2 \alpha - \sin^2 \alpha) = -(\cos^2 \alpha + \sin^2 \alpha) \\ = -1$$

Now,

$$A_{11} = -\cos^2 \alpha - \sin^2 \alpha = -1 \quad A_{12} = 0 \quad A_{13} = 0$$

$$A_{21} = 0 \quad A_{22} = -\cos \alpha \quad A_{23} = -\sin \alpha$$

$$A_{31} = 0 \quad A_{32} = -\sin \alpha \quad A_{33} = \cos \alpha$$

Therefore,

$$\text{adj } A = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -\cos \alpha & -\sin \alpha \\ 0 & -\sin \alpha & \cos \alpha \end{bmatrix}$$

Hence,

Conclusion

$$A^{-1} = \frac{1}{|A|} \text{adj } A \\ = -1 \begin{bmatrix} -1 & 0 & 0 \\ 0 & -\cos \alpha & -\sin \alpha \\ 0 & -\sin \alpha & \cos \alpha \end{bmatrix} \\ = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \alpha & \sin \alpha \\ 0 & \sin \alpha & -\cos \alpha \end{bmatrix}$$

Q

30

Verifying
 $(AB)^{-1} = B^{-1} A^{-1}$



Let $A = \begin{bmatrix} 3 & 7 \\ 2 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 6 & 8 \\ 7 & 9 \end{bmatrix}$. Verify that $(AB)^{-1} = B^{-1} A^{-1}$.

Given



$$\text{Let } A = \begin{bmatrix} 3 & 7 \\ 2 & 5 \end{bmatrix}$$

Then,

$$|A| = 15 - 14 \\ = 1$$

Now,

$$A_{11} = 5 \quad A_{12} = -2$$

$$A_{21} = -7 \quad A_{22} = 3$$

Then,

$$\text{adj } A = \begin{bmatrix} 5 & -7 \\ -2 & 3 \end{bmatrix}$$

Therefore,

$$A^{-1} = \frac{1}{|A|} \text{adj } A$$

$$= \begin{pmatrix} 5 & -7 \\ -2 & 3 \end{pmatrix}$$

Substitute

Now,

$$\text{Let } B = \begin{pmatrix} 6 & 8 \\ 7 & 9 \end{pmatrix}$$

Then,

$$|B| = 54 - 56$$

$$= -2$$

Now,

$$A_{11} = 9 \quad A_{12} = -7$$

$$A_{21} = -8 \quad A_{22} = 6$$

Then,

$$\text{adj } B = \begin{pmatrix} 9 & -8 \\ -7 & 6 \end{pmatrix}$$

Therefore,

$$B^{-1} = \frac{1}{|B|} \text{adj } B$$

$$= -\frac{1}{2} \begin{pmatrix} 9 & -8 \\ -7 & 6 \end{pmatrix}$$

$$= \begin{pmatrix} -\frac{9}{2} & 4 \\ \frac{7}{2} & -3 \end{pmatrix}$$

Now,

$$B^{-1} A^{-1} = \begin{pmatrix} -\frac{9}{2} & 4 \\ \frac{7}{2} & -3 \end{pmatrix} \begin{pmatrix} 5 & -7 \\ -2 & 3 \end{pmatrix}$$

$$= \begin{pmatrix} -\frac{45}{2} - 8 & \frac{63}{2} + 12 \\ \frac{35}{2} + 6 & -\frac{49}{2} - 9 \end{pmatrix}$$

$$= \begin{pmatrix} -\frac{61}{2} & \frac{87}{2} \\ \frac{47}{2} & -\frac{67}{2} \end{pmatrix}$$

Also,

$$A B = \begin{pmatrix} 3 & 7 \\ 2 & 5 \end{pmatrix} \begin{pmatrix} 6 & 8 \\ 7 & 9 \end{pmatrix}$$

$$= \begin{pmatrix} 18+49 & 24+63 \\ 12+35 & 16+45 \end{pmatrix}$$

$$= \begin{pmatrix} 67 & 87 \\ 47 & 61 \end{pmatrix}$$

Then, we have

$$\begin{aligned}
 |A B| &= 67(61) - 87(47) \\
 &= 4087 - 4089 \\
 &= -2
 \end{aligned}$$

Therefore,

$$\text{adj}(A B) = \begin{pmatrix} 61 & -87 \\ -47 & 67 \end{pmatrix}$$

Thus,

$$\begin{aligned}
 (A B)^{-1} &= \frac{1}{|A B|} \text{adj}(A B) \\
 &= -\frac{1}{2} \begin{pmatrix} 61 & -87 \\ -47 & 67 \end{pmatrix} \\
 &= \begin{pmatrix} -\frac{61}{2} & \frac{87}{2} \\ \frac{47}{2} & -\frac{67}{2} \end{pmatrix}
 \end{aligned}$$

Conclusion

From (1) and (2),

$$(A B)^{-1} = B^{-1} A^{-1}$$

Hence, proved.

Q

31

Matrix
polynomial
identity and
inverse via it



If $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$, show that $A^2 - 5A + 7I = O$. Hence find A^{-1} .

Given

$$\text{Let } A = \begin{pmatrix} 3 & 1 \\ -1 & 2 \end{pmatrix}$$

Substitute

Therefore,

$$\begin{aligned}
 A^2 &= A \cdot A = \begin{pmatrix} 3 & 1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 3 & 1 \\ -1 & 2 \end{pmatrix} \\
 &= \begin{pmatrix} 9-1 & 3+2 \\ -3-2 & -1+4 \end{pmatrix} \\
 &= \begin{pmatrix} 8 & 5 \\ -5 & 3 \end{pmatrix}
 \end{aligned}$$

Now,

$$\begin{aligned}
 A^2 - 5A + 7I &= \begin{pmatrix} 8 & 5 \\ -5 & 3 \end{pmatrix} - 5 \begin{pmatrix} 3 & 1 \\ -1 & 2 \end{pmatrix} + 7 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\
 &= \begin{pmatrix} 8 & 5 \\ -5 & 3 \end{pmatrix} - \begin{pmatrix} 15 & 5 \\ -5 & 10 \end{pmatrix} + \begin{pmatrix} 7 & 0 \\ 0 & 7 \end{pmatrix} \\
 &= \begin{pmatrix} -7 & 0 \\ 0 & -7 \end{pmatrix} + \begin{pmatrix} 7 & 0 \\ 0 & 7 \end{pmatrix} \\
 &= \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}
 \end{aligned}$$

Hence, $A^2 - 5A + 7I = O$.

Now,

$$\begin{aligned}
&\Rightarrow A \cdot A - 5A = -7I \\
&\Rightarrow A \cdot A(A^{-1}) - 5A \cdot A^{-1} = -7IA^{-1} \quad [\text{post-multiplying by } A^{-1} \text{ as } |A| \neq 0] \\
&\Rightarrow A(AA^{-1}) - 5I = -7A^{-1} \\
&\Rightarrow AI - 5I = -7A^{-1} \\
&\Rightarrow A^{-1} = -\frac{1}{7}(A - 5I) \\
&\Rightarrow A^{-1} = \frac{1}{7}(5I - A) \\
&\Rightarrow A^{-1} = \frac{1}{7} \left[\begin{pmatrix} 5 & 0 \\ 0 & 5 \end{pmatrix} - \begin{pmatrix} 3 & 1 \\ -1 & 2 \end{pmatrix} \right] \\
&\Rightarrow A^{-1} = \frac{1}{7} \begin{pmatrix} 2 & -1 \\ 1 & 3 \end{pmatrix}
\end{aligned}$$

Thus,

Conclusion

$$A^{-1} = \frac{1}{7} \begin{pmatrix} 2 & -1 \\ 1 & 3 \end{pmatrix}$$

Q

32

Finding a, b
from a
matrix
polynomial
identity

(())

For the matrix $A = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$, find the numbers a and b such that $A^2 + aA + bI = O$.

Given

$$\text{Let } A = \begin{pmatrix} 3 & 2 \\ 1 & 1 \end{pmatrix}$$

Substitute

Therefore,

$$\begin{aligned}
A^2 &= A \cdot A = \begin{pmatrix} 3 & 2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 3 & 2 \\ 1 & 1 \end{pmatrix} \\
&= \begin{pmatrix} 9+2 & 6+2 \\ 3+1 & 2+1 \end{pmatrix} = \begin{pmatrix} 11 & 8 \\ 4 & 3 \end{pmatrix}
\end{aligned}$$

Now, $A^2 + aA + bI = O$.

Hence,

$$\begin{aligned}
&\Rightarrow (A \cdot A)A^{-1} + aA \cdot A^{-1} + bIA^{-1} = O \quad [\text{post-multiplying by } A^{-1} \text{ as } |A| \neq 0] \\
&\Rightarrow A(AA^{-1}) + aI + b(AA^{-1}) = O \\
&\Rightarrow AI + aI + bA^{-1} = O \\
&\Rightarrow A + aI = -bA^{-1} \\
&\Rightarrow A^{-1} = -\frac{1}{b}(A + aI) \quad \dots(1)
\end{aligned}$$

Now,

$$\begin{aligned}
A^{-1} &= \frac{1}{|A|} \text{adj } A \\
&= \frac{1}{1} \begin{pmatrix} 1 & -2 \\ -1 & 3 \end{pmatrix} \\
&= \begin{pmatrix} 1 & -2 \\ -1 & 3 \end{pmatrix}
\end{aligned}$$

From (1) and (2), we have,

$$\Rightarrow \begin{pmatrix} 1 & -2 \\ -1 & 3 \end{pmatrix} = \frac{1}{b} \left[\begin{pmatrix} 3 & 2 \\ 1 & 1 \end{pmatrix} + \begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix} \right]$$

$$\Rightarrow \begin{pmatrix} 1 & -2 \\ -1 & 3 \end{pmatrix} = -\frac{1}{b} \begin{pmatrix} 3+a & 2 \\ 1 & a \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 1 & -2 \\ -1 & 3 \end{pmatrix} = \begin{pmatrix} \frac{-3-a}{b} & -\frac{2}{b} \\ -\frac{1}{b} & \frac{-1-a}{b} \end{pmatrix}$$

Comparing the corresponding elements of the two matrices, we have:

$$\Rightarrow -\frac{1}{b} = -1$$

$$\Rightarrow b = 1$$

Also,

$$\Rightarrow \frac{-3-a}{b} = 1$$

$$\Rightarrow -3-a = 1$$

$$\Rightarrow a = -4$$

Conclusion

Thus, $a = -4$ and $b = 1$.

Q

33

Matrix
polynomial
identity and
inverse via it
(3x3)



For the matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix}$ Show that $A^3 - 6A^2 + 5A + 11I = O$. Hence, find A^{-1} .

2-11

Given

Let

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{pmatrix}$$

Substitute

Therefore,

$$A^2 = A \cdot A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{pmatrix}$$

$$= \begin{pmatrix} 1+1+2 & 1+2-1 & 1-3+3 \\ 1+2-6 & 1+4+3 & 1-6-9 \\ 2-1+6 & 2-2-3 & 2+3+9 \end{pmatrix} = \begin{pmatrix} 4 & 2 & 1 \\ -3 & 8 & -14 \\ 7 & -3 & 14 \end{pmatrix}$$

And,

$$A^3 = A^2 \cdot A = \begin{pmatrix} 4 & 2 & 1 \\ -3 & 8 & -14 \\ 7 & -3 & 14 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{pmatrix}$$

$$= \begin{pmatrix} 4+2+2 & 4+4-1 & 4-6+3 \\ -3+8-28 & -3+16+14 & -3-24-42 \\ 7-3+28 & 7-6-14 & 7+9+42 \end{pmatrix} = \begin{pmatrix} 8 & 7 & 1 \\ -23 & 27 & -69 \\ 32 & -13 & 58 \end{pmatrix}$$

Hence,

$$\begin{aligned}
A^3 - 6A^2 + 5A + 11I &= \begin{pmatrix} 8 & 7 & 1 \\ -23 & 27 & -69 \\ 32 & -13 & 58 \end{pmatrix} - 6 \begin{pmatrix} 4 & 2 & 1 \\ -3 & 8 & -14 \\ 7 & -3 & 14 \end{pmatrix} + 5 \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{pmatrix} + 11 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\
&= \begin{pmatrix} 8 & 7 & 1 \\ -23 & 27 & -69 \\ 32 & -13 & 58 \end{pmatrix} - \begin{pmatrix} 24 & 12 & 6 \\ -18 & 48 & -84 \\ 42 & -18 & 84 \end{pmatrix} + \begin{pmatrix} 5 & 5 & 5 \\ 5 & 10 & -15 \\ 10 & -5 & 15 \end{pmatrix} + \begin{pmatrix} 11 & 0 & 0 \\ 0 & 11 & 0 \\ 0 & 0 & 11 \end{pmatrix} \\
&= \begin{pmatrix} 24 & 12 & 6 \\ -18 & 48 & -84 \\ 42 & -18 & 84 \end{pmatrix} - \begin{pmatrix} 24 & 12 & 6 \\ -18 & 48 & -84 \\ 42 & -18 & 84 \end{pmatrix} \\
&= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\
&= 0
\end{aligned}$$

Thus, $A^3 - 6A^2 + 5A + 11I = 0$

Now,

$$\begin{aligned}
&\Rightarrow A^3 - 6A^2 + 5A + 11I = 0 \\
&\Rightarrow (A A A) A^{-1} - 6(A A) A^{-1} + 5A A^{-1} + 11I A^{-1} = 0 \quad [\text{post-multiplying by } A^{-1} \text{ as } |A| \neq 0] \\
&\Rightarrow A A(A A^{-1}) - 6A(A A^{-1}) + 5(A A^{-1}) = -11(I A^{-1}) \\
&\Rightarrow A^2 - 6A + 5I = -11A^{-1} \\
&\Rightarrow A^{-1} = -\frac{1}{11}(A^2 - 6A + 5I)
\end{aligned}$$

Now,

$$\begin{aligned}
A^2 - 6A + 5I &= \begin{pmatrix} 4 & 2 & 1 \\ -3 & 8 & -14 \\ 7 & -3 & 14 \end{pmatrix} - 6 \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{pmatrix} + 5 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\
&= \begin{pmatrix} 4 & 2 & 1 \\ -3 & 8 & -14 \\ 7 & -3 & 14 \end{pmatrix} - \begin{pmatrix} 6 & 6 & 6 \\ 6 & 12 & -18 \\ 12 & -6 & 18 \end{pmatrix} + \begin{pmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{pmatrix} \\
&= \begin{pmatrix} 9 & 2 & 1 \\ -3 & 13 & -14 \\ 7 & -3 & 19 \end{pmatrix} - \begin{pmatrix} 6 & 6 & 6 \\ 6 & 12 & -18 \\ 12 & -6 & 18 \end{pmatrix} \\
&= \begin{pmatrix} 3 & -4 & -5 \\ -9 & 1 & 4 \\ -5 & 3 & 1 \end{pmatrix}
\end{aligned}$$

From equation (1) and (2)

Conclusion

$$\begin{aligned}
A^{-1} &= -\frac{1}{11} \begin{pmatrix} 3 & -4 & -5 \\ -9 & 1 & 4 \\ -5 & 3 & 1 \end{pmatrix} \\
&= \frac{1}{11} \begin{pmatrix} -3 & 4 & 5 \\ 9 & -1 & -4 \\ 5 & -3 & -1 \end{pmatrix}
\end{aligned}$$

Q

34

Matrix
polynomial
identity and
inverse via it
(3x3)



If $A = \begin{bmatrix} -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$ Verify that $A^3 - 6A^2 + 9A - 4I = O$ and hence find A^{-1}

Given

Let

$$A = \begin{pmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{pmatrix}$$

Substitute

Therefore,

$$\begin{aligned}
A^2 &= A \cdot A \\
&= \begin{pmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{pmatrix} \begin{pmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{pmatrix} \\
&= \begin{pmatrix} 4+1+1 & -2-2-1 & 2+1+2 \\ -2-2-1 & 1+4+1 & -1-2-2 \\ 2+1+2 & -1-2-2 & 1+1+4 \end{pmatrix} \\
&= \begin{pmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{pmatrix}
\end{aligned}$$

And

$$\begin{aligned}
A^3 &= A^2 \cdot A \\
&= \begin{pmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{pmatrix} \begin{pmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{pmatrix} \\
&= \begin{pmatrix} 12+5+5 & -6-10-5 & 6+5+10 \\ -10-6-5 & 5+12+5 & -5-6-10 \\ 10+5+6 & -5-10-6 & 5+5+12 \end{pmatrix} \\
&= \begin{pmatrix} 22 & -21 & 21 \\ -21 & 22 & -21 \\ 21 & -21 & 22 \end{pmatrix}
\end{aligned}$$

Now,

$$\begin{aligned}
A^3 - 6A^2 + 9A - 4I &= \begin{pmatrix} 22 & -21 & 21 \\ -21 & 22 & -21 \\ 21 & -21 & 22 \end{pmatrix} - 6 \begin{pmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{pmatrix} + 9 \begin{pmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{pmatrix} - 4 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\
&= \begin{pmatrix} 22 & -21 & 21 \\ -21 & 22 & -21 \\ 21 & -21 & 22 \end{pmatrix} - \begin{pmatrix} 36 & -30 & 30 \\ -30 & 36 & -30 \\ 30 & -30 & 36 \end{pmatrix} + \begin{pmatrix} 18 & -9 & 9 \\ -9 & 18 & -9 \\ 9 & -9 & 18 \end{pmatrix} - \begin{pmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{pmatrix} \\
&= \begin{pmatrix} 40 & -30 & 30 \\ -30 & 40 & -30 \\ 30 & -30 & 40 \end{pmatrix} - \begin{pmatrix} 40 & -30 & 30 \\ -30 & 40 & -30 \\ 30 & -30 & 40 \end{pmatrix} \\
&= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\
&= 0
\end{aligned}$$

Thus,

$$A^3 - 6A^2 + 9A - 4I = 0$$

Now,

$$\begin{aligned}
&\Rightarrow A^3 - 6A^2 + 9A - 4I = 0 \\
&\Rightarrow (A \ A \ A) A^{-1} - 6(A \ A) A^{-1} + 9A A^{-1} - 4I A^{-1} = 0 \quad [\text{post-multiplying by } A^{-1} \text{ as } |A| \neq 0] \\
&\Rightarrow A \ A (A A^{-1}) - 6A (A A^{-1}) + 9(A A^{-1}) = 4(I A^{-1}) \\
&\Rightarrow A \ A \ I - 6A \ I + 9I = 4A^{-1} \\
&\Rightarrow A^2 - 6A + 9I = 4A^{-1} \\
&\Rightarrow A^{-1} = \frac{1}{4} (A^2 - 6A + 9I)
\end{aligned}$$

Now,

$$\begin{aligned}
 A^2 - 6A + 9I &= \begin{pmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{pmatrix} - 6 \begin{pmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{pmatrix} + 9 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\
 &= \begin{pmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{pmatrix} - \begin{pmatrix} 12 & -6 & 6 \\ -6 & 12 & -6 \\ 6 & -6 & 12 \end{pmatrix} + \begin{pmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{pmatrix} \\
 &= \begin{pmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 3 \end{pmatrix}
 \end{aligned}$$

From equations (1) and (2),

Conclusion

$$A^{-1} = \frac{1}{4} \begin{pmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 3 \end{pmatrix}$$

Q

35

MCQ - $|\text{adj } A|$ for a nonsingular 3×3 matrix



Let A be a nonsingular square matrix of order 3×3 . Then $|\text{adj } A|$ is equal to (A) $|A|$ (B) $|A|^2$ (C) $|A|^3$ (D) $3|A|$

Given

Since A is a non-singular square matrix of order 3×3

Substitute

$$\begin{aligned}
 (\text{adj } A) A &= |A| I \\
 &= \begin{pmatrix} |A| & 0 & 0 \\ 0 & |A| & 0 \\ 0 & 0 & |A| \end{pmatrix}
 \end{aligned}$$

Therefore,

Conclusion

$$\begin{aligned}
 |(\text{adj } A) A| &= \begin{vmatrix} |A| & 0 & 0 \\ 0 & |A| & 0 \\ 0 & 0 & |A| \end{vmatrix} \\
 |\text{adj } A| |A| &= |A|^3 \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} \\
 &= |A|^3 I \\
 |\text{adj } A| &= |A|^2
 \end{aligned}$$

Thus, the correct option is B.

Q

36

MCQ - $\det(A^{-1})$ for an invertible 2×2 matrix

(())

If A is an invertible matrix of order 2, then $\det(A^{-1})$ is equal to (A) $\det(A)$ (B)

$\frac{1}{\det(A)}$ (C) 1 (D) 0

Given

Since A is an invertible matrix, A^{-1} exists.

$$\text{Also, } A^{-1} = \frac{1}{|A|} \text{adj } A.$$

$$\text{As matrix } A \text{ is of order 2, let } A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

Substitute

Then,

$$|A| = a d - b c$$

And

$$\text{adj } A = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Now,

$$\begin{aligned} A^{-1} &= \frac{1}{|A|} \text{adj } A \\ &= \begin{pmatrix} \frac{d}{|A|} & \frac{-b}{|A|} \\ \frac{-c}{|A|} & \frac{a}{|A|} \end{pmatrix} \end{aligned}$$

Hence,

$$\begin{aligned} |A^{-1}| &= \begin{vmatrix} \frac{d}{|A|} & \frac{-b}{|A|} \\ \frac{-c}{|A|} & \frac{a}{|A|} \end{vmatrix} \\ |A^{-1}| &= \frac{1}{|A|^2} \begin{vmatrix} d & -b \\ -c & a \end{vmatrix} \\ &= \frac{1}{|A|^2} (a d - b c) \\ &= \frac{1}{|A|^2} \cdot |A| \\ &= \frac{1}{|A|} \end{aligned}$$

Hence,

Conclusion

$$\det(A^{-1}) = \frac{1}{\det(A)}$$

Thus, the correct option is B.

Q

37

**Examining
consistency
of a linear
system**



Examine the consistency of the system of equations in Exercises 1 to 6.

$$\begin{aligned} x + 2y &= 2 \\ 2x + 3y &= 3 \end{aligned}$$

Given

$$x + 2y = 2$$

The given system of equations is: $2x + 3y = 3$

We can write the given system of equations in the form $A X = B$, where

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}, X = \begin{bmatrix} x \\ y \end{bmatrix} \text{ and } B = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

Substitute

Hence,

$$\begin{aligned} |A| &= 1(3) - 2(2) \\ &= 3 - 4 \\ &= -1 \\ &\neq 0 \end{aligned}$$

Conclusion

So, A is non-singular.

Therefore, A^{-1} exists.

Thus, the given system of equations is consistent.

Q

38

Examining
consistency
of a linear
system



Examine the consistency of the system of equations in Exercises 1 to 6.

$$\begin{aligned}2x - y &= 5 \\ x + y &= 4\end{aligned}$$

Given

$$2x - y = 5$$

The given system of equations is: $x + y = 4$

We can write the given system of equations in the form $AX = B$, where

$$A = \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}, X = \begin{bmatrix} x \\ y \end{bmatrix} \text{ and } B = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$$

Substitute

Hence,

$$\begin{aligned}|A| &= 2(1) - 1(-1) \\ &= 2 + 1 \\ &= 3 \\ &\neq 0\end{aligned}$$

Conclusion

So, A is non-singular.

Therefore, A^{-1} exists.

Hence, the given system of equations is consistent.

Q

39

Examining
consistency
of a linear
system



Examine the consistency of the system of equations in Exercises 1 to 6.

$$\begin{aligned}x + 3y &= 5 \\ 2x + 6y &= 8\end{aligned}$$

Given

$$x + 3y = 5$$

The given system of equations is: $2x + 6y = 8$

We can write the given system of equations in the form $AX = B$, where

$$A = \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix}, X = \begin{bmatrix} x \\ y \end{bmatrix} \text{ and } B = \begin{bmatrix} 5 \\ 8 \end{bmatrix}$$

Substitute

Hence,

$$\begin{aligned}|A| &= 1(6) - 3(2) \\ &= 6 - 6 \\ &= 0\end{aligned}$$

So, A is a singular matrix.

Now,

$$(\text{adj } A) = \begin{bmatrix} 6 & -3 \\ -2 & 1 \end{bmatrix}$$

Therefore,

$$\begin{aligned}
 (\text{adj } A) B &= \begin{pmatrix} 6 & -5 \\ -2 & 1 \end{pmatrix} \begin{bmatrix} 5 \\ 8 \end{bmatrix} \\
 &= \begin{bmatrix} 30-24 \\ -10+8 \end{bmatrix} \\
 &= \begin{bmatrix} 6 \\ -2 \end{bmatrix} \\
 &\neq 0
 \end{aligned}$$

Conclusion

Thus, the solution of the given system of equations does not exist.

Hence, the system of equations is inconsistent.

Q

40

Examining consistency of a linear system with a parameter

(())

Examine the consistency of the system of equations in Exercises 1 to 6.

$$\begin{aligned}
 x+y+z &= 1 \\
 2x+3y+2z &= 2 \\
 ax+ay+2az &= 4
 \end{aligned}$$

Given

$$\begin{aligned}
 x+y+z &= 1 \\
 2x+3y+2z &= 2
 \end{aligned}$$

The given system of equations is: $ax+ay+2az = 4$

We can write the given system of equations in the form $AX = B$, where

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 3 & 2 \\ a & a & 2a \end{pmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}$$

Substitute

Hence,

$$\begin{aligned}
 |A| &= 1(6a-2a)-1(4a-2a)+1(2a-3a) \\
 &= 4a-2a-a \\
 &= 4a-3a \\
 &= a \neq 0
 \end{aligned}$$

Conclusion

So, A is non-singular.

Therefore, A^{-1} exists.

Thus, the given system of equations is consistent.

Q

41

Examining consistency of a linear system

(())

Examine the consistency of the system of equations in Exercises 1 to 6.

$$\begin{aligned}
 3x-y-2z &= 2 \\
 2y-z &= -1 \\
 3x-5y &= 3
 \end{aligned}$$

Given

$$\begin{aligned}
 3x-y-2z &= 2 \\
 2y-z &= -1
 \end{aligned}$$

The given system of equations is: $3x-5y = 3$

We can write the given system of equations in the form $AX = B$, where

$$A = \begin{pmatrix} 3 & -1 & -2 \\ 0 & 2 & -1 \\ 3 & -5 & 0 \end{pmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } B = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$$

Substitute

Hence,

$$\begin{aligned}
 |A| &= 3(0-5)-0+3(1+4) \\
 &= -15+15 \\
 &= 0
 \end{aligned}$$

So, A is a singular matrix.

Now,

$$(\text{adj } A) = \begin{bmatrix} -5 & 10 & 5 \\ -3 & 6 & 3 \\ -6 & 12 & 6 \end{bmatrix}$$

Therefore,

$$\begin{aligned}
 (\text{adj } A) B &= \begin{bmatrix} -5 & 10 & 5 \\ -3 & 6 & 3 \\ -6 & 12 & 6 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} \\
 &= \begin{bmatrix} -10-10+15 \\ -6-6+9 \\ -12-12+18 \end{bmatrix} \\
 &= \begin{bmatrix} -5 \\ -3 \\ -6 \end{bmatrix} \\
 &\neq 0
 \end{aligned}$$

Conclusion

Thus, the solution of the given system of equations does not exist.

Hence, the system of equations is inconsistent.

Q

42

Examining
consistency
of a linear
system



Examine the consistency of the system of equations in Exercises 1 to 6.

$$\begin{aligned}
 5x - y + 4z &= 5 \\
 2x + 3y + 5z &= 2 \\
 5x - 2y + 6z &= -1
 \end{aligned}$$

Given



$$\begin{aligned}
 5x - y + 4z &= 5 \\
 2x + 3y + 5z &= 2
 \end{aligned}$$

The given system of equations is: $5x - 2y + 6z = -1$

We can write the given system of equations in the form $AX = B$, where

$$A = \begin{bmatrix} 5 & -1 & 4 \\ 2 & 3 & 5 \\ 5 & -2 & 6 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } B = \begin{bmatrix} 5 \\ 2 \\ -1 \end{bmatrix}$$

Substitute

Hence,

$$\begin{aligned}
 |A| &= 5(18+10)+1(12-25)+4(-4-15) \\
 &= 5(28)+1(-13)+4(-19) \\
 &= 140-13-76 \\
 &= 51 \neq 0
 \end{aligned}$$

Conclusion

So, A is nonsingular.

Therefore, A^{-1} exists.

Hence, the given system of equations is consistent.

Q**43**

Solving a
linear system
by matrix
method



Solve system of linear equations, using matrix method, in Exercises 7 to 14.

$$\begin{aligned} 5x + 2y &= 4 \\ 7x + 3y &= 5 \end{aligned}$$

Given

$$5x + 2y = 4$$

The given system of equations is: $7x + 3y = 5$

We can write the given system of equations in the form $AX = B$, where

$$A = \begin{pmatrix} 5 & 2 \\ 7 & 3 \end{pmatrix}, X = \begin{bmatrix} x \\ y \end{bmatrix} \text{ and } B = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

Substitute

Hence,

$$\begin{aligned} |A| &= 15 - 14 \\ &= 1 \\ &\neq 0 \end{aligned}$$

So, A is non-singular.

Therefore, A^{-1} exists.

Key formula

Now,

$$\begin{aligned} A^{-1} &= \frac{1}{|A|} (\text{adj } A) \\ &= \begin{pmatrix} 3 & -2 \\ -7 & 5 \end{pmatrix} \end{aligned}$$

Substitute

Then,

$$\begin{aligned} \Rightarrow X &= A^{-1} B \\ \Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} &= \begin{pmatrix} 3 & -2 \\ -7 & 5 \end{pmatrix} \begin{bmatrix} 4 \\ 5 \end{bmatrix} \\ \Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} &= \begin{bmatrix} 12 - 10 \\ -28 + 25 \end{bmatrix} \\ \Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} &= \begin{bmatrix} 2 \\ -3 \end{bmatrix} \end{aligned}$$

Conclusion

Hence, $x = 2$ and $y = -3$

Q**44**

Solving a
linear system
by matrix
method

(())

Solve system of linear equations, using matrix method, in Exercises 7 to 14.

$$\begin{aligned} 2x - y &= -2 \\ 3x + 4y &= 3 \end{aligned}$$

Given

$$2x - y = -2$$

The given system of equations is: $3x + 4y = 3$

We can write the given system of equations in the form $AX = B$, where

$$A = \begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix}, X = \begin{bmatrix} x \\ y \end{bmatrix} \text{ and } B = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$$

Substitute

Hence,

$$\begin{aligned} |A| &= 8 + 3 \\ &= 11 \\ &\neq 0 \end{aligned}$$

So, A is non-singular.

Therefore, A^{-1} exists.

Key formula

Now,

$$A^{-1} = \frac{1}{|A|} (\text{adj } A)$$

$$= \frac{1}{11} \begin{pmatrix} 4 & 1 \\ -3 & 2 \end{pmatrix}$$

Substitute

Therefore,

$$\Rightarrow X = A^{-1} B$$

$$\Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{11} \begin{pmatrix} 4 & 1 \\ -3 & 2 \end{pmatrix} \begin{bmatrix} -2 \\ 3 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{11} \begin{bmatrix} -8+3 \\ 6+6 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{11} \begin{bmatrix} -5 \\ 12 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \frac{-5}{11} \\ \frac{12}{11} \end{bmatrix}$$

Conclusion

Hence, $x = \frac{-5}{11}$ and $y = \frac{12}{11}$

Q

45

Solving a
linear system
by matrix
method



Solve system of linear equations, using matrix method, in Exercises 7 to 14.

$$4x - 3y = 3$$

$$3x - 5y = 7$$

Given

$$4x - 3y = 3$$

The given system of equations is: $3x - 5y = 7$

We can write the given system of equations in the form $AX = B$, where

$$A = \begin{pmatrix} 4 & -3 \\ 3 & -5 \end{pmatrix}, X = \begin{bmatrix} x \\ y \end{bmatrix} \text{ and } B = \begin{bmatrix} 3 \\ 7 \end{bmatrix}$$

Substitute

Hence,

$$|A| = -20 + 9$$

$$= -11$$

$$\neq 0$$

So, A is nonsingular.

Therefore, A^{-1} exists.

Now,

Key formula

$$A^{-1} = \frac{1}{|A|} (\text{adj } A)$$

$$= -\frac{1}{11} \begin{pmatrix} -5 & 3 \\ -3 & 4 \end{pmatrix} = \frac{1}{11} \begin{pmatrix} 5 & -3 \\ 3 & -4 \end{pmatrix}$$

Substitute

Therefore,

$$\Rightarrow X = A^{-1} B$$

$$\Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{11} \begin{pmatrix} 5 & -3 \\ 3 & -4 \end{pmatrix} \begin{bmatrix} 3 \\ 7 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{11} \begin{pmatrix} 5 & -3 \\ 3 & -4 \end{pmatrix} \begin{bmatrix} 3 \\ 7 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{11} \begin{bmatrix} 15-21 \\ 9-28 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{11} \begin{bmatrix} -6 \\ -19 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \frac{-6}{11} \\ \frac{-19}{11} \end{bmatrix}$$

Conclusion

Hence, $x = \frac{-6}{11}$ and $y = \frac{-19}{11}$

Q

46

Solving a linear system by matrix method



Solve system of linear equations, using matrix method, in Exercises 7 to 14.

$$\begin{aligned} 5x + 2y &= 3 \\ 3x + 2y &= 5 \end{aligned}$$

Given

$$5x + 2y = 3$$

The given system of equations is: $3x + 2y = 5$

We can write the given system of equations in the form $AX = B$, where

$$A = \begin{pmatrix} 5 & 2 \\ 3 & 2 \end{pmatrix}, X = \begin{bmatrix} x \\ y \end{bmatrix} \text{ and } B = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

Substitute

Hence,

$$\begin{aligned} |A| &= 10 - 6 \\ &= 4 \\ &\neq 0 \end{aligned}$$

So, A is non-singular.

Therefore, A^{-1} exists.

Now,

Key formula

$$\begin{aligned} A^{-1} &= \frac{1}{|A|} (\text{adj } A) \\ &= \frac{1}{4} \begin{pmatrix} 2 & -2 \\ -3 & 5 \end{pmatrix} \end{aligned}$$

Substitute

Therefore,

$$\Rightarrow X = A^{-1} B$$

$$\Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{4} \begin{pmatrix} 2 & -2 \\ -3 & 5 \end{pmatrix} \begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{4} \begin{pmatrix} 2 & -2 \\ -3 & 5 \end{pmatrix} \begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 6-10 \\ -9+25 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{4} \begin{bmatrix} -4 \\ 16 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$$

Conclusion

Hence, $x = -1$ and $y = 4$

Q

47

Solving a 3-variable linear system by matrix method



Solve system of linear equations, using matrix method, in Exercises 7 to 14.

$$2x + y + z = 1$$

$$x - 2y - z = \frac{3}{2}$$

$$3y - 5z = 9$$

Given

$$2x + y + z = 1$$

$$x - 2y - z = \frac{3}{2}$$

The given system of equations is: $3y - 5z = 9$

We can write the given system of equations in the form $AX = B$, where

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 1 & -2 & -1 \\ 0 & 3 & -5 \end{pmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 \\ \frac{3}{2} \\ 9 \end{bmatrix}$$

Substitute

Hence,

$$\begin{aligned} |A| &= 2(10+3) - 1(-5-3) + 0 \\ &= 2(13) - 1(-8) \\ &= 26 + 8 \\ &= 34 \\ &\neq 0 \end{aligned}$$

So, A is non-singular.

Therefore, A^{-1} exists.

Now,

$$\begin{aligned} A_{11} &= 13 \quad A_{12} = 5 \quad A_{13} = 3 \\ A_{21} &= 8 \quad A_{22} = -10 \quad A_{23} = -6 \\ A_{31} &= 1 \quad A_{32} = 3 \quad A_{33} = -5 \end{aligned}$$

Hence,

Key formula

$$\begin{aligned} A^{-1} &= \frac{1}{|A|} (\text{adj } A) \\ &= \frac{1}{34} \begin{pmatrix} 13 & 8 & 1 \\ 5 & -10 & 3 \\ 3 & -6 & -5 \end{pmatrix} \end{aligned}$$

Substitute

Therefore,

$$\Rightarrow X = A^{-1} B$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{34} \begin{bmatrix} 13 & 8 & 1 \\ 5 & -10 & 3 \\ 3 & -6 & -5 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{34} \begin{bmatrix} 13 & 8 & 1 \\ 5 & -10 & 3 \\ 3 & -6 & -5 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{34} \begin{bmatrix} 13+12+9 \\ 5-15+27 \\ 3-9-45 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{34} \begin{bmatrix} 34 \\ 17 \\ -51 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Conclusion

$$\text{Hence, } x = 1, y = \frac{1}{2} \text{ and } z = \frac{-3}{2}$$

Q

48

Solving a 3-variable linear system by matrix method

(())

Solve system of linear equations, using matrix method, in Exercises 7 to 14.

$$\begin{aligned} x-y+z &= 4 \\ 2x+y-3z &= 0 \\ x+y+z &= 2 \end{aligned}$$

Given

$$\begin{aligned} x-y+z &= 4 \\ 2x+y-3z &= 0 \end{aligned}$$

The given system of equations is: $x+y+z = 2$

We can write the given system of equations in the form $AX = B$, where $A =$

$$\begin{bmatrix} 1 & -1 & 1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } B = \begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix}$$

Substitute

Hence,

$$\begin{aligned} |A| &= 1(1+3)+1(2+3)+1(2-1) \\ &= 4+5+1 \\ &= 10 \\ &\neq 0 \end{aligned}$$

So, A is nonsingular.

Therefore, A^{-1} exists.

Now,

$$\begin{aligned} A_{11} &= 4, A_{12} = -5, A_{13} = 1 \\ A_{21} &= 2, A_{22} = 0, A_{23} = -2 \\ A_{31} &= 1, A_{32} = 1, A_{33} = 1 \end{aligned}$$

Hence,

Key formula

$$A^{-1} = \frac{1}{|A|} (\text{adj } A)$$

$$= \frac{1}{10} \begin{pmatrix} 4 & 2 & 2 \\ -5 & 0 & 5 \\ 1 & -2 & 3 \end{pmatrix}$$

Substitute

Therefore,

$$\Rightarrow X = A^{-1} B$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{10} \begin{pmatrix} 4 & 2 & 2 \\ -5 & 0 & 5 \\ 1 & -2 & 3 \end{pmatrix} \begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{10} \begin{pmatrix} 4 & 2 & 2 \\ -5 & 0 & 5 \\ 1 & -2 & 3 \end{pmatrix} \begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{10} \begin{bmatrix} 16+0+4 \\ -20+0+10 \\ 4+0+6 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{10} \begin{bmatrix} 20 \\ -10 \\ 10 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

Conclusion

Hence, $x = 2$, $y = -1$ and $z = 1$

Q**49**

Solving a 3-variable linear system by matrix method



Solve system of linear equations, using matrix method, in Exercises 7 to 14.

$$\begin{aligned} 2x+3y+3z &= 5 \\ x-2y+z &= -4 \\ 3x-y-2z &= 3 \end{aligned}$$

Given

$$\begin{aligned} 2x+3y+3z &= 5 \\ x-2y+z &= -4 \end{aligned}$$

The given system of equations is: $3x - y - 2z = 3$

We can write the given system of equations in the form $AX = B$, where

$$A = \begin{pmatrix} 2 & 3 & 3 \\ 1 & -2 & 1 \\ 3 & -1 & -2 \end{pmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } B = \begin{bmatrix} 5 \\ -4 \\ 3 \end{bmatrix}$$

Substitute

Hence,

$$\begin{aligned} |A| &= 2(4+1)-3(-2-3)+3(-1+6) \\ &= 10+15+15 \\ &= 40 \\ &\neq 0 \end{aligned}$$

So, A is non-singular.

Therefore, A^{-1} exists.

Now,

$$\begin{aligned} A_{11} &= 5 \quad A_{12} = 5 \quad A_{13} = 5 \\ A_{21} &= 3 \quad A_{22} = -13 \quad A_{23} = 11 \\ A_{31} &= 9 \quad A_{32} = 1 \quad A_{33} = -7 \end{aligned}$$

Hence,

Key formula

$$A^{-1} = \frac{1}{|A|} (\text{adj } A)$$

$$= \frac{1}{40} \begin{pmatrix} 5 & 3 & 9 \\ 5 & -13 & 1 \\ 5 & 11 & -7 \end{pmatrix}$$

Substitute

Therefore,

$$\Rightarrow X = A^{-1} B$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{40} \begin{pmatrix} 5 & 3 & 9 \\ 5 & -13 & 1 \\ 5 & 11 & -7 \end{pmatrix} \begin{bmatrix} 5 \\ -4 \\ 3 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{40} \begin{pmatrix} 5 & 3 & 9 \\ 5 & -13 & 1 \\ 5 & 11 & -7 \end{pmatrix} \begin{bmatrix} 5 \\ -4 \\ 3 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{40} \begin{bmatrix} 25-12+27 & 25+52+3 & 25-44-21 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{40} \begin{bmatrix} 40 \\ 80 \\ -40 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

Conclusion

Hence, $x = 1$, $y = 2$ and $z = -1$

Q**50**

**Solving a 3-
variable
linear system
by matrix
method**



Solve system of linear equations, using matrix method, in Exercises 7 to 14.

$$\begin{aligned} x-y+2z &= 7 \\ 3x+4y-5z &= -5 \\ 2x-y+3z &= 12 \end{aligned}$$

Given

$$\begin{aligned} x-y+2z &= 7 \\ 3x+4y-5z &= -5 \end{aligned}$$

The given system of equations is: $2x-y+3z = 12$

We can write the given system of equations in the form $AX = B$, where

$$A = \begin{pmatrix} 1 & -1 & 2 \\ 3 & 4 & -5 \\ 2 & -1 & 3 \end{pmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } B = \begin{bmatrix} 7 \\ -5 \\ 12 \end{bmatrix}$$

Substitute

Hence,

$$\begin{aligned} |A| &= 1(12-5)+1(9+10)+2(-3-8) \\ &= 7+19-22 \\ &= 4 \\ &\neq 0 \end{aligned}$$

So, A is non-singular.

Therefore, A^{-1} exists.

Now,

$$\begin{aligned} A_{11} &= 7 \quad A_{12} = -19 \quad A_{13} = -11 \\ A_{21} &= 1 \quad A_{22} = -1 \quad A_{23} = -1 \\ A_{31} &= -3 \quad A_{32} = 11 \quad A_{33} = 7 \end{aligned}$$

Hence,

Key formula

$$A^{-1} = \frac{1}{|A|} (\text{adj } A)$$

$$= \frac{1}{4} \begin{pmatrix} 7 & 1 & -3 \\ -19 & -1 & 11 \\ -11 & -1 & 7 \end{pmatrix}$$

Substitute

Therefore,

$$\Rightarrow X = A^{-1} B$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{4} \begin{pmatrix} 7 & 1 & -3 \\ -19 & -1 & 11 \\ -11 & -1 & 7 \end{pmatrix} \begin{bmatrix} 7 \\ -5 \\ 12 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{4} \begin{pmatrix} 7 & 1 & -3 \\ -19 & -1 & 11 \\ -11 & -1 & 7 \end{pmatrix} \begin{bmatrix} 7 \\ -5 \\ 12 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 49-5-36 \\ -133+5+132 \\ -77+5+84 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 49-5-36 \\ -133+5+132 \\ -77+5+84 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 8 \\ 4 \\ 12 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$$

Conclusion

Hence, $x = 2$, $y = 1$ and $z = 3$

Q**51**

**Solving a
system using
a given
inverse
matrix**



If $A = \begin{bmatrix} 2 & -3 & 5 \\ 3 & 2 & -4 \\ 1 & 1 & -2 \end{bmatrix}$, find A^{-1} . Using A^{-1} solve the system of equations

$$\begin{aligned} 2x - 3y + 5z &= 11 \\ 3x + 2y - 4z &= -5 \\ x + y - 2z &= -3 \end{aligned}$$

Given

It is given that

$$A = \begin{pmatrix} 2 & -3 & 5 \\ 3 & 2 & -4 \\ 1 & 1 & -2 \end{pmatrix}$$

Substitute

Therefore,

$$\begin{aligned} |A| &= 2(-4+4)+3(-6+4)+5(3-2) \\ &= 0-6+5 \\ &= -1 \\ &\neq 0 \end{aligned}$$

Now,

$$\begin{aligned} A_{11} &= 0 \quad A_{12} = 2 \quad A_{13} = 1 \\ A_{21} &= -1 \quad A_{22} = -9 \quad A_{23} = -5 \\ A_{31} &= 2 \quad A_{32} = 23 \quad A_{33} = 13 \end{aligned}$$

Hence,

Key formula

$$A^{-1} = \frac{1}{|A|} (\text{adj } A)$$

$$= - \begin{pmatrix} 0 & -1 & 2 \\ 2 & -9 & 23 \\ 1 & -5 & 13 \end{pmatrix} = \begin{pmatrix} 0 & 1 & -2 \\ -2 & 9 & -23 \\ -1 & 5 & -13 \end{pmatrix}$$

Given

We can write the given system of equations in the form $A X = B$, where

$$A = \begin{pmatrix} 2 & -3 & 5 \\ 3 & 2 & -4 \\ 1 & 1 & -2 \end{pmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } B = \begin{bmatrix} 11 \\ -5 \\ -3 \end{bmatrix}$$

The solution of the system of equations is $X = A^{-1} B$.

Substitute

Therefore,

$$\Rightarrow X = A^{-1} B$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{pmatrix} 0 & 1 & -2 \\ -2 & 9 & -23 \\ -1 & 5 & -13 \end{pmatrix} \begin{bmatrix} 11 \\ -5 \\ -3 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{pmatrix} 0 & 1 & -2 \\ -2 & 9 & -23 \\ -1 & 5 & -13 \end{pmatrix} \begin{bmatrix} 11 \\ -5 \\ -3 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0-5+6 \\ -22-45+69 \\ -11-25+39 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Conclusion

Hence, $x = 1$, $y = 2$ and $z = 3$

Q**52**

Word
problem -
cost per kg
via matrix
method

(())

The cost of 4 kg onion, 3 kg wheat and 2 kg rice is ₹ 60 . The cost of 2 kg onion, 4 kg wheat and 6 kg rice is ₹ 90 . The cost of 6 kg onion 2 kg wheat and 3 kg rice is ₹ 70. Find cost of each item per kg by matrix method.

Given

Let the cost of onions, wheat, and rice per kg in ₹ be x , y and z respectively.

We can write this situation as a system of equations:

$$4x + 3y + 2z = 60$$

$$2x + 4y + 6z = 90$$

$$6x + 2y + 3z = 70$$

We can write the given system of equations in the form $A X = B$, where

$$A = \begin{pmatrix} 4 & 3 & 2 \\ 2 & 4 & 6 \\ 6 & 2 & 3 \end{pmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } B = \begin{bmatrix} 60 \\ 90 \\ 70 \end{bmatrix}$$

Substitute

Therefore,

$$|A| = 4(12-12)-3(6-36)+2(4-24)$$

$$= 0+90-40$$

$$= 50$$

$$\neq 0$$

So, A is non-singular.

Therefore, A^{-1} exists.

Now,

$$\begin{aligned} A_{11} &= 0 \quad A_{12} = 30 \quad A_{13} = -20 \\ A_{21} &= -5 \quad A_{22} = 0 \quad A_{23} = 10 \\ A_{31} &= 10 \quad A_{32} = -20 \quad A_{33} = 10 \end{aligned}$$

Therefore,

Key formula



$$\begin{aligned} A^{-1} &= \frac{1}{|A|} (\text{adj } A) \\ &= \frac{1}{50} \begin{pmatrix} 0 & -5 & 10 \\ 30 & 0 & -20 \\ -20 & 10 & 10 \end{pmatrix} \end{aligned}$$

Substitute

Hence,

$$\begin{aligned} \Rightarrow X &= A^{-1} B \\ \Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \frac{1}{50} \begin{pmatrix} 0 & -5 & 10 \\ 30 & 0 & -20 \\ -20 & 10 & 10 \end{pmatrix} \begin{bmatrix} 60 \\ 90 \\ 70 \end{bmatrix} \\ \Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \frac{1}{50} \begin{pmatrix} 0 & -5 & 10 \\ 30 & 0 & -20 \\ -20 & 10 & 10 \end{pmatrix} \begin{bmatrix} 60 \\ 90 \\ 70 \end{bmatrix} \\ \Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \frac{1}{50} \begin{bmatrix} 0-450+700 \\ 1800+0-1400 \\ -1200+900+700 \end{bmatrix} \\ \Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \frac{1}{50} \begin{bmatrix} 250 \\ 400 \\ 400 \end{bmatrix} \\ \Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \begin{bmatrix} 5 \\ 8 \\ 8 \end{bmatrix} \end{aligned}$$

Conclusion

Thus, $x = 5$, $y = 8$ and $z = 8$

Hence, the cost of onions is ₹ 5 per kg the cost of wheat is ₹ 8 per kg, and the cost of rice is ₹ 8 per kg.

Q

53

Determinant independent of theta



Prove that the determinant

$$\begin{vmatrix} x & \sin \theta & \cos \theta \\ -\sin \theta & -x & 1 \\ \cos \theta & 1 & x \end{vmatrix} \text{ is independent of } \theta.$$

Given



$$\begin{aligned} \Delta &= \begin{vmatrix} x & \sin \theta & \cos \theta \\ -\sin \theta & -x & 1 \\ \cos \theta & 1 & x \end{vmatrix} \\ &= x(-x^2-1)-\sin \theta(-x \sin \theta-\cos \theta)+\cos \theta(-\sin \theta+x \cos \theta) \\ &= -x^3-x+x \sin^2 \theta+\sin \theta \cos \theta-\sin \theta \cos \theta+x \cos^2 \theta \\ &= -x^3-x+x(\sin^2 \theta+\cos^2 \theta) \\ &= -x^3-x+x \\ &= -x^3 \end{aligned}$$

Conclusion

Hence, Δ is independent of θ .

Q

54

Evaluating a
determinant
with trig
entries



Evaluate

$$\begin{vmatrix} \cos \alpha \cos \beta & \cos \alpha \sin \beta & -\sin \alpha \\ -\sin \beta & \cos \beta & 0 \\ \sin \alpha \cos \beta & \sin \alpha \sin \beta & \cos \alpha \end{vmatrix}.$$

Conclusion

Let

$$\Delta = \begin{vmatrix} \cos \alpha \cos \beta & \cos \alpha \sin \beta & -\sin \alpha \\ -\sin \beta & \cos \beta & 0 \\ \sin \alpha \cos \beta & \sin \alpha \sin \beta & \cos \alpha \end{vmatrix}$$

Expanding along C_3 ,

$$\begin{aligned} \Delta &= -\sin \alpha (-\sin \alpha \sin^2 \beta - \cos^2 \beta \sin \alpha) + \cos \alpha (\cos \alpha \cos^2 \beta + \cos \alpha \sin^2 \beta) \\ &= \sin^2 \alpha (\sin^2 \beta + \cos^2 \beta) + \cos^2 \alpha (\cos^2 \beta + \sin^2 \beta) \\ &= \sin^2 \alpha (1) + \cos^2 \alpha (1) \\ &= 1 \end{aligned}$$

Q

55

Finding
 $(AB)^{-1}$ from
 A^{-1} and B



If $A^{-1} = \begin{bmatrix} 3 & -1 & 1 \\ -15 & 6 & -5 \\ 5 & -2 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & -2 \\ -1 & 3 & 0 \\ 0 & -2 & 1 \end{bmatrix}$, find $(AB)^{-1}$

Key formula

We know that $(A B)^{-1} = B^{-1} A^{-1}$.

Substitute

It is given that

$$B = \begin{bmatrix} 1 & 2 & -2 \\ -1 & 3 & 0 \\ 0 & -2 & 1 \end{bmatrix}$$

Therefore,

$$\begin{aligned} |B| &= 1(3) - 2(-1) - 2(-2) \\ &= 3 + 2 - 4 \\ &= 5 - 4 \\ &= 1 \end{aligned}$$

Now,

$$\begin{aligned} B_{11} &= 3 \quad B_{12} = 1 \quad B_{13} = 2 \\ B_{21} &= 2 \quad B_{22} = 1 \quad B_{23} = 2 \\ B_{31} &= 6 \quad B_{32} = 2 \quad B_{33} = 5 \end{aligned}$$

Hence,

$$\text{adj } B = \begin{pmatrix} 3 & 2 & 6 \\ 1 & 1 & 2 \\ 2 & 2 & 5 \end{pmatrix}$$

Now,

$$\begin{aligned} B^{-1} &= \frac{1}{|B|} \text{adj } B \\ &= \begin{pmatrix} 3 & 2 & 6 \\ 1 & 1 & 2 \\ 2 & 2 & 5 \end{pmatrix} \end{aligned}$$

Therefore,

$$\begin{aligned}
 (A B)^{-1} &= B^{-1} A^{-1} \\
 &= \begin{pmatrix} 3 & 2 & 6 \\ 1 & 1 & 2 \\ 2 & 2 & 5 \end{pmatrix} \begin{pmatrix} 3 & -1 & 1 \\ -15 & 6 & -5 \\ 5 & -2 & 2 \end{pmatrix} \\
 &= \begin{pmatrix} 9-30+30 & -3+12-12 & 3-10+12 \\ 3-15+10 & -1+6-4 & 1-5+4 \\ 6-30+25 & -2+12-10 & 2-10+10 \end{pmatrix} \\
 &= \begin{pmatrix} 9 & -3 & 5 \\ -2 & 1 & 0 \\ 1 & 0 & 2 \end{pmatrix}
 \end{aligned}$$

Conclusion

Thus,

$$(A B)^{-1} = \begin{pmatrix} 9 & -3 & 5 \\ -2 & 1 & 0 \\ 1 & 0 & 2 \end{pmatrix}.$$

Q

56

Verifying
adjoint/inverse
identities

(())

Let $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & 1 \\ 1 & 1 & 5 \end{bmatrix}$. Verify that

Part (i)

$$[\text{adj } A]^{-1} = \text{adj } (A^{-1})$$

Part (ii)

$$(A^{-1})^{-1} = A$$

Given

$$A = \begin{pmatrix} 1 & -2 & 1 \\ -2 & 3 & 1 \\ 1 & 1 & 5 \end{pmatrix}$$

Therefore,

$$\begin{aligned}
 |A| &= 1(15-1)+2(-10-1)+1(-2-3) \\
 &= 14-22-5 \\
 &= -13
 \end{aligned}$$

Now,

$$\begin{aligned}
 A_{11} &= 14 \quad A_{12} = 11 \quad A_{13} = -5 \\
 A_{21} &= 11 \quad A_{22} = 4 \quad A_{23} = -3 \\
 A_{31} &= -5 \quad A_{32} = -3 \quad A_{33} = -1
 \end{aligned}$$

Hence,

$$\text{adj } A = \begin{pmatrix} 14 & 11 & -5 \\ 11 & 4 & -3 \\ -5 & -3 & -1 \end{pmatrix}$$

Now,

$$\begin{aligned}
 A^{-1} &= \frac{1}{|A|} (\text{adj } A) \\
 &= -\frac{1}{13} \begin{pmatrix} 14 & 11 & -5 \\ 11 & 4 & -3 \\ -5 & -3 & -1 \end{pmatrix} \\
 &= \frac{1}{13} \begin{pmatrix} -14 & -11 & 5 \\ -11 & -4 & 3 \\ 5 & 3 & 1 \end{pmatrix}
 \end{aligned}$$

Substitute

(i)

$$\begin{aligned}
 |\text{adj } A| &= 14(-4-9)-11(-11-15)-5(-33+20) \\
 &= 14(-13)-11(-26)-5(-13) \\
 &= -182+286+65 \\
 &= 169
 \end{aligned}$$

We have,

$$\text{adj}(\text{adj } A) = \begin{pmatrix} -13 & 26 & -13 \\ 26 & -39 & -13 \\ -13 & -13 & -65 \end{pmatrix}$$

Therefore,

$$\begin{aligned}
 [\text{adj } A]^{-1} &= \frac{1}{|\text{adj } A|} (\text{adj}(\text{adj } A)) \\
 &= \frac{1}{169} \begin{pmatrix} -13 & 26 & -13 \\ 26 & -39 & -13 \\ -13 & -13 & -65 \end{pmatrix} \\
 &= \begin{pmatrix} \frac{-1}{13} & \frac{2}{13} & \frac{-1}{13} \\ \frac{2}{13} & \frac{-3}{13} & \frac{-1}{13} \\ \frac{-1}{13} & \frac{-1}{13} & \frac{-5}{13} \end{pmatrix}
 \end{aligned}$$

Now,

$$A^{-1} = \frac{1}{13} \begin{pmatrix} 14 & 11 & -5 \\ 11 & 4 & -3 \\ -5 & -3 & -1 \end{pmatrix} = \begin{pmatrix} \frac{-14}{13} & \frac{-11}{13} & \frac{5}{13} \\ \frac{-11}{13} & \frac{-4}{13} & \frac{3}{13} \\ \frac{5}{13} & \frac{3}{13} & \frac{1}{13} \end{pmatrix}$$

Therefore,

$$\begin{aligned}
 \text{adj}(A)^{-1} &= \begin{pmatrix} \frac{-13}{169} & \frac{26}{169} & \frac{-13}{169} \\ \frac{26}{169} & \frac{-39}{169} & \frac{-13}{169} \\ \frac{-13}{169} & \frac{-13}{169} & \frac{-65}{169} \end{pmatrix} \\
 &= \begin{pmatrix} \frac{-1}{13} & \frac{2}{13} & \frac{-1}{13} \\ \frac{2}{13} & \frac{-3}{13} & \frac{-1}{13} \\ \frac{-1}{13} & \frac{-1}{13} & \frac{-5}{13} \end{pmatrix}
 \end{aligned}$$

Hence, $[\text{adj } A]^{-1} = \text{adj}(A)^{-1}$ proved.

Conclusion

(ii)

$$A^{-1} = \frac{1}{13} \begin{pmatrix} -14 & -11 & 5 \\ -11 & -4 & 3 \\ 5 & 3 & 1 \end{pmatrix}$$

Hence,

$$\text{adj}(A)^{-1} = \begin{bmatrix} \frac{-1}{13} & \frac{2}{13} & \frac{-1}{13} \\ \frac{2}{13} & \frac{-3}{13} & \frac{-1}{13} \\ \frac{-1}{13} & \frac{-1}{13} & \frac{-5}{13} \end{bmatrix}$$

Now,

$$\begin{aligned} |A^{-1}| &= \left(\frac{1}{13} \right)^3 [-14(-4-9)+11(-11-26)+5(-33+20)] \\ &= \left(\frac{1}{13} \right)^3 [-169] \\ &= -\frac{1}{13} \end{aligned}$$

Therefore,

$$\begin{aligned} (A^{-1})^{-1} &= \frac{\text{adj } A^{-1}}{|A|} = \frac{1}{\left(-\frac{1}{13} \right)} \times \begin{bmatrix} \frac{-1}{13} & \frac{2}{13} & \frac{-1}{13} \\ \frac{2}{13} & \frac{-3}{13} & \frac{-1}{13} \\ \frac{-1}{13} & \frac{-1}{13} & \frac{-5}{13} \end{bmatrix} \\ &= \begin{bmatrix} 1 & -2 & 1 \\ -2 & 3 & 1 \\ 1 & 1 & 5 \end{bmatrix} = A \end{aligned}$$

Hence, $(A^{-1})^{-1} = A$ proved.

Q

57

Evaluating a
symmetric-
sum
determinant



Evaluate

$$\begin{vmatrix} x & y & x+y \\ y & x+y & x \\ x+y & x & y \end{vmatrix}$$

Conclusion

$$\begin{aligned} \Delta &= \begin{vmatrix} x & y & x+y \\ y & x+y & x \\ x+y & x & y \end{vmatrix} \\ &= \begin{vmatrix} 2(x+y) & 2(x+y) & 2(x+y) \\ y & x+y & x \\ x+y & x & y \end{vmatrix} \quad [R_1 \rightarrow R_1 + R_2 + R_3] \\ &= 2(x+y) \begin{vmatrix} 1 & 1 & 1 \\ y & x+y & x \\ x+y & x & y \end{vmatrix} \\ &= 2(x+y) \begin{vmatrix} 1 & 0 & 0 \\ y & x & x-y \\ x+y & -y & -x \end{vmatrix} \quad [C_2 \rightarrow C_2 - C_1 \text{ and } C_3 \rightarrow C_3 - C_1] \\ &= 2(x+y)[-x^2 + y(x-y)] \quad [\text{Expanding along } R_1] \\ &= -2(x+y)(x^2 + y^2 - yx) \\ &= -2(x^3 + y^3) \end{aligned}$$

Q

58

Evaluating a determinant with x, y entries



Evaluate

$$\begin{vmatrix} 1 & x & y \\ 1 & x+y & y \\ 1 & x & x+y \end{vmatrix}$$

Conclusion

$$\begin{aligned} \Delta &= \begin{vmatrix} 1 & x & y \\ 1 & x+y & y \\ 1 & x & x+y \end{vmatrix} \\ &= \begin{vmatrix} 1 & x & y \\ 0 & y & 0 \\ 0 & 0 & x \end{vmatrix} \\ &= 1(x \cdot y - 0) [R_2 \rightarrow R_2 - R_1 \text{ and } R_3 \rightarrow R_3 - R_1] \\ &= x y [\text{Expanding along } C_1] \end{aligned}$$

Q

59

Solving a system of reciprocal equations



Solve the system of equations

$$\begin{aligned} \frac{2}{x} + \frac{3}{y} + \frac{10}{z} &= 4 \\ \frac{4}{x} - \frac{6}{y} + \frac{5}{z} &= 1 \\ \frac{6}{x} + \frac{9}{y} - \frac{20}{z} &= 2 \end{aligned}$$

Choose the correct answer in Exercise 17 to 19 .

Given

Let $\frac{1}{x} = p$, $\frac{1}{y} = q$ and $\frac{1}{z} = r$.

Then the given system of equations is as follows:

$$\begin{aligned} 2p + 3q + 10r &= 4 \\ 4p - 6q + 5r &= 1 \\ 6p + 9q - 20r &= 2 \end{aligned}$$

This system can be written in the form of $A X = B$, where

$$A = \begin{pmatrix} 2 & 3 & 10 \\ 4 & -6 & 5 \\ 6 & 9 & -20 \end{pmatrix}, X = \begin{bmatrix} p \\ q \\ r \end{bmatrix} \text{ and } B = \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}$$

Substitute

Therefore,

$$\begin{aligned} |A| &= 2(120 - 45) - 3(-80 - 30) + 10(36 + 36) \\ &= 150 + 330 + 720 \\ &= 1200 \end{aligned}$$

Thus, A is non-singular.

Therefore, A^{-1} exists.

Now,

$$\begin{aligned} A_{11} &= 75 \quad A_{12} = 110 \quad A_{13} = 72 \\ A_{21} &= 150 \quad A_{22} = -100 \quad A_{23} = 0 \\ A_{31} &= 75 \quad A_{32} = 30 \quad A_{33} = -24 \end{aligned}$$

Hence,

$$\begin{aligned} A^{-1} &= \frac{1}{|A|} (\text{adj } A) \\ &= \frac{1}{1200} \begin{pmatrix} 75 & 150 & 75 \\ 110 & -100 & 30 \\ 72 & 0 & -24 \end{pmatrix} \end{aligned}$$

Now,

$$\Rightarrow X = A^{-1} B$$

$$\Rightarrow \begin{bmatrix} p \\ q \\ r \end{bmatrix} = \frac{1}{1200} \begin{bmatrix} 75 & 150 & 75 \\ 110 & -100 & 30 \\ 72 & 0 & -24 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} p \\ q \\ r \end{bmatrix} = \frac{1}{1200} \begin{bmatrix} 300+150+150 \\ 440-100+60 \\ 288+0-48 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} p \\ q \\ r \end{bmatrix} = \frac{1}{1200} \begin{bmatrix} 600 \\ 400 \\ 240 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} p \\ q \\ r \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{3} \\ \frac{1}{5} \end{bmatrix}$$

Conclusion

Therefore,

$$p = \frac{1}{2}, q = \frac{1}{3} \text{ and } r = \frac{1}{5}$$

Hence, $x = 2$, $y = 3$ and $z = 5$.

Q

60

MCQ -
inverse of a
diagonal
matrix

(())

Choose the correct answer in Exercise 17 to 19. If x, y, z are nonzero real

numbers, then the inverse of matrix $A = \begin{bmatrix} x & 0 & 0 \\ 0 & y & 0 \\ 0 & 0 & z \end{bmatrix}$ is (A) $\begin{bmatrix} x^{-1} & 0 & 0 \\ 0 & y^{-1} & 0 \\ 0 & 0 & z^{-1} \end{bmatrix}$ (B) $x y z$

(C) $\frac{1}{x y z} \begin{bmatrix} x & 0 & 0 \\ 0 & y & 0 \\ 0 & 0 & z \end{bmatrix}$ (D) $\frac{1}{x y z} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Given

It is given that

$$A = \begin{bmatrix} x & 0 & 0 \\ 0 & y & 0 \\ 0 & 0 & z \end{bmatrix}$$

Hence,

$$\begin{aligned} |A| &= x(y z - 0) \\ &= x y z \\ &\neq 0 \end{aligned}$$

Substitute

Now,

$$\begin{aligned} A_{11} &= y z \quad A_{12} = 0 \quad A_{13} = 0 \\ A_{21} &= 0 \quad A_{22} = x z \quad A_{23} = 0 \\ A_{31} &= 0 \quad A_{32} = 0 \quad A_{33} = x y \end{aligned}$$

Therefore,

$$\begin{aligned}
A^{-1} &= \frac{1}{|A|} (\text{adj } A) \\
&= \frac{1}{xyz} \begin{bmatrix} yz & 0 & 0 \\ 0 & xz & 0 \\ 0 & 0 & xy \end{bmatrix} \\
&= \begin{bmatrix} \frac{yz}{xyz} & 0 & 0 \\ 0 & \frac{xz}{xyz} & 0 \\ 0 & 0 & \frac{xy}{xyz} \end{bmatrix} \\
&= \begin{bmatrix} \frac{1}{x} & 0 & 0 \\ 0 & \frac{1}{y} & 0 \\ 0 & 0 & \frac{1}{z} \end{bmatrix} \\
&= \begin{bmatrix} x^{-1} & 0 & 0 \\ 0 & y^{-1} & 0 \\ 0 & 0 & z^{-1} \end{bmatrix}
\end{aligned}$$

Conclusion

Thus, the correct option is A.

Q

61

MCQ - range
of a
determinant
with trig
entries



Choose the correct answer in Exercise 17 to 19. Let $A = \begin{bmatrix} 1 & \sin \theta & 1 \\ -\sin \theta & 1 & \sin \theta \\ -1 & -\sin \theta & 1 \end{bmatrix}$, where $0 \leq \theta \leq 2\pi$. Then (A) $\text{Det}(A) = 0$ (B) $\text{Det}(A) \in (2, \infty)$ (C) $\text{Det}(A) \in (2, 4)$ (D) $\text{Det}(A) \in [2, 4]$

Given

It is given that

$$A = \begin{bmatrix} 1 & \sin \theta & 1 \\ -\sin \theta & 1 & \sin \theta \\ -1 & -\sin \theta & 1 \end{bmatrix}$$

Hence,

$$\begin{aligned}
|A| &= 1(1 + \sin^2 \theta) - \sin \theta(-\sin \theta + \sin \theta) + 1(\sin^2 \theta + 1) \\
&= 1 + \sin^2 \theta + \sin^2 \theta + 1 \\
&= 2 + 2 \sin^2 \theta \\
&= 2(1 + \sin^2 \theta)
\end{aligned}$$

Substitute

Now,

$$\begin{aligned}
&\Rightarrow 0 \leq \theta \leq 2\pi \\
&\Rightarrow -1 \leq \sin \theta \leq 1 \\
&\Rightarrow 0 \leq \sin^2 \theta \leq 1 \\
&\Rightarrow 1 \leq 1 + \sin^2 \theta \leq 2 \\
&\Rightarrow 2 \leq 2(1 + \sin^2 \theta) \leq 4
\end{aligned}$$

Conclusion

Therefore,

$$\text{Det}(A) \in [2, 4]$$

Thus, the correct option is D.

Q

1

Properties of determinants



Using the property of determinants and without expanding, prove that:

$$\begin{vmatrix} x & a & x+a \\ y & b & y+b \\ z & c & z+c \end{vmatrix} = 0$$

Key formula



$$\begin{aligned} \Delta &= \begin{vmatrix} x & a & x+a \\ y & b & y+b \\ z & c & z+c \end{vmatrix} \\ &= \begin{vmatrix} x & a & x \\ y & b & y \\ z & c & z \end{vmatrix} + \begin{vmatrix} x & a & a \\ y & b & b \\ z & c & c \end{vmatrix} \end{aligned}$$

Identical columns

Here, two columns of each determinant are identical.

Conclusion

Hence,

$$\begin{aligned} \Delta &= 0+0 \\ &= 0 \end{aligned}$$

Q

2

Properties of determinants



Using the property of determinants and without expanding, prove that:

$$\begin{vmatrix} a-b & b-c & c-a \\ b-c & c-a & a-b \\ c-a & a-b & b-c \end{vmatrix} = 0$$

Key formula



$$\begin{aligned} \Delta &= \begin{vmatrix} a-b & b-c & c-a \\ b-c & c-a & a-b \\ c-a & a-b & b-c \end{vmatrix} \\ &= \begin{vmatrix} a-c & b-a & c-b \\ b-c & c-a & a-b \\ -(a-c) & -(b-a) & -(c-b) \end{vmatrix} \\ &= \begin{vmatrix} a-c & b-a & c-b \\ b-c & c-a & a-b \\ a-c & b-a & c-b \end{vmatrix} \end{aligned}$$

Identical rows

Here, the two rows R_1 and R_3 are identical.

Conclusion

Hence, $\Delta = 0$

Q

3

Properties of determinants



Using the property of determinants and without expanding, prove that:

$$\begin{vmatrix} 2 & 7 & 65 \\ 3 & 8 & 75 \\ 5 & 9 & 86 \end{vmatrix} = 0$$

Key formula

$$\begin{aligned}
 \Delta &= \begin{vmatrix} 2 & 7 & 65 \\ 3 & 8 & 75 \\ 5 & 9 & 86 \end{vmatrix} \\
 &= \begin{vmatrix} 2 & 7 & 63+2 \\ 3 & 8 & 72+3 \\ 5 & 9 & 81+5 \end{vmatrix} \\
 &= \begin{vmatrix} 2 & 7 & 63 \\ 3 & 8 & 72 \\ 5 & 9 & 81 \end{vmatrix} + \begin{vmatrix} 2 & 7 & 2 \\ 3 & 8 & 3 \\ 5 & 9 & 5 \end{vmatrix} \quad [\because \text{Two columns are identical}] \\
 &= \begin{vmatrix} 2 & 7 & 9(7) \\ 3 & 8 & 9(8) \\ 5 & 9 & 9(9) \end{vmatrix} + 0 \\
 &= 9 \begin{vmatrix} 2 & 7 & 7 \\ 3 & 8 & 8 \\ 5 & 9 & 9 \end{vmatrix} \\
 &= 0
 \end{aligned}$$

Q**4****Properties of determinants****Using the property of determinants and without expanding, prove that:**

$$\begin{vmatrix} 1 & b & c & a(b+c) \\ 1 & c & a & b(c+a) \\ 1 & a & b & c(a+b) \end{vmatrix} = 0$$

Key formula

$$\begin{aligned}
 \Delta &= \begin{vmatrix} 1 & b & c & a(b+c) \\ 1 & c & a & b(c+a) \\ 1 & a & b & c(a+b) \end{vmatrix} \\
 &= \begin{vmatrix} 1 & b & c & a+b+c+a \\ 1 & c & a & a+b+c+a \\ 1 & a & b & a+b+c+a \end{vmatrix} \quad [C_4 \rightarrow C_4 + C_3]
 \end{aligned}$$

Proportional columnsHere, the two columns C_1 and C_3 are proportional.**Conclusion**Hence, $\Delta = 0$ **Q****5****Properties of determinants****Using the property of determinants and without expanding, prove that:**

$$\begin{vmatrix} b+c & q+r & y+z \\ c+a & r+p & z+x \\ a+b & p+q & x+y \end{vmatrix} = 2 \begin{vmatrix} a & p & x \\ b & q & y \\ c & r & z \end{vmatrix}$$

Key formula

$$\begin{aligned}
 \Delta &= \begin{vmatrix} b+c & q+r & y+z \\ c+a & r+p & z+x \\ a+b & p+q & x+y \end{vmatrix} \\
 &= \begin{vmatrix} b+c & q+r & y+z \\ c+a & r+p & z+x \\ a & p & x \end{vmatrix} + \begin{vmatrix} b+c & q+r & y+z \\ c+a & r+p & z+x \\ b & q & y \end{vmatrix} \\
 &= \Delta_1 + \Delta_2
 \end{aligned}$$

Substitute

Now,

$$\begin{aligned}
 \Delta_1 &= \begin{vmatrix} b+c & q+r & y+z \\ c+a & r+p & z+x \\ a & p & x \end{vmatrix} \\
 &= \begin{vmatrix} b+c & q+r & y+z \\ c & r & z \\ a & p & x \end{vmatrix} [R_2 \rightarrow R_2 - R_3] \\
 &= \begin{vmatrix} b & q & y \\ c & r & z \\ a & p & x \end{vmatrix} \\
 &= (-1)^2 \begin{vmatrix} a & p & x \\ b & q & y \\ c & r & z \end{vmatrix} [R_1 \rightarrow R_1 - R_2] \\
 \Delta_1 &= \begin{vmatrix} a & p & x \\ b & q & y \\ c & r & z \end{vmatrix} [R_1 \leftrightarrow R_3 \text{ and } R_2 \leftrightarrow R_3] \\
 \Delta_2 &= \begin{vmatrix} b+c & q+r & y+z \\ c+a & r+p & z+x \\ b & q & y \end{vmatrix} [R_1 \rightarrow R_1 - R_3] \\
 \Delta_2 &= \begin{vmatrix} c & r & z \\ c+a & r+p & z+x \\ b & q & y \end{vmatrix} [R_2 \rightarrow R_2 - R_1] \\
 \Delta_2 &= \begin{vmatrix} c & r & z \\ a & p & x \\ b & q & y \end{vmatrix} \\
 \Delta_2 &= (-1)^2 \begin{vmatrix} a & p & x \\ b & q & y \\ c & r & z \end{vmatrix} [R_1 \leftrightarrow R_2 \text{ and } R_2 \leftrightarrow R_3]
 \end{aligned}$$

Conclusion

From (1),(2) and (3), we have

$$\begin{aligned}
 \Delta &= \begin{vmatrix} a & p & x \\ b & q & y \\ c & r & z \end{vmatrix} + \begin{vmatrix} a & p & x \\ b & q & y \\ c & r & z \end{vmatrix} \\
 &= 2 \begin{vmatrix} a & p & x \\ b & q & y \\ c & r & z \end{vmatrix} \\
 \text{Hence, } \begin{vmatrix} b+c & q+r & y+z \\ c+a & r+p & z+x \\ a+b & p+q & x+y \end{vmatrix} &= 2 \begin{vmatrix} a & p & x \\ b & q & y \\ c & r & z \end{vmatrix} \text{ proved.}
 \end{aligned}$$

Q

6

Properties of
determinants



Using the property of determinants and without expanding, prove that:

$$\begin{vmatrix} 0 & a & -b \\ -a & 0 & -c \\ b & c & 0 \end{vmatrix} = 0$$

Key formula

$$\begin{aligned}
 \Delta &= \begin{vmatrix} 0 & a & -b \\ -a & 0 & -c \\ b & c & 0 \end{vmatrix} \\
 &= \frac{1}{c} \begin{vmatrix} 0 & a & -b \\ -a & 0 & -c \\ b & c & 0 \end{vmatrix} [R_1 \rightarrow c R_1] \\
 &= \frac{1}{c} \begin{vmatrix} a & b & -c \\ -a & 0 & -c \\ b & c & 0 \end{vmatrix} [R_1 \rightarrow R_1 - b R_2] \\
 &= \frac{a}{c} \begin{vmatrix} b & c & 0 \\ -a & 0 & -c \\ b & c & 0 \end{vmatrix}
 \end{aligned}$$

Identical rows

Here, the two rows R_1 and R_3 are identical.

Conclusion

Hence, $\Delta = 0$

Q

7

Properties of determinants

(i)

Using the property of determinants and without expanding, prove that:

$$\begin{vmatrix} -a^2 & ab & ac \\ ba & -b^2 & bc \\ ca & cb & -c^2 \end{vmatrix} = 4a^2b^2c^2$$

Key formula

$$\begin{aligned} \Delta &= \begin{vmatrix} -a^2 & ab & ac \\ ba & -b^2 & bc \\ ca & cb & -c^2 \end{vmatrix} \\ &= abc \begin{vmatrix} -a & b & c \\ a & -b & c \\ a & b & -c \end{vmatrix} \quad [\text{Taking out factors } a, b, c \text{ from } R_1, R_2, R_3] \\ &= a^2b^2c^2 \begin{vmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix} \quad [\text{Taking out factors } a, b, c \text{ from } C_1, C_2, C_3] \\ &= a^2b^2c^2 \begin{vmatrix} -1 & 1 & 1 \\ 0 & 0 & 2 \\ 0 & 2 & 0 \end{vmatrix} \quad [R_2 \rightarrow R_2 + R_1 \text{ and } R_3 \rightarrow R_3 + R_1] \\ &= a^2b^2c^2(-1) \begin{vmatrix} 0 & 2 \\ 2 & 0 \end{vmatrix} \\ &= -a^2b^2c^2(0-4) \\ &= 4a^2b^2c^2 \end{aligned}$$

Q

8

Properties of determinants

(ii)

By using properties of determinants show that: (i)

$$\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = (a-b)(b-c)(c-a)$$

$$(ii) \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(a+b+c)$$

Given

$$\begin{aligned} (i) \text{ Let } \Delta &= \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} \\ \Delta &= \begin{vmatrix} 0 & a-c & a^2-c^2 \\ 0 & b-c & b^2-c^2 \\ 1 & c & c^2 \end{vmatrix} \quad [R_1 \rightarrow R_1 - R_3 \text{ and } R_2 \rightarrow R_2 - R_3] \\ &= (c-a)(b-c) \begin{vmatrix} 0 & -1 & -a-c \\ 0 & 1 & b+c \\ 1 & c & c^2 \end{vmatrix} \\ &= (b-c)(c-a) \begin{vmatrix} 0 & 0 & -a+b \\ 0 & 1 & b+c \\ 1 & c & c^2 \end{vmatrix} \quad [R_1 \rightarrow R_1 + R_2] \quad = (a-b)(b-c)(c-a) \\ &= (a-b)(b-c)(c-a) \begin{vmatrix} 0 & 0 & -1 \\ 0 & 1 & b+c \\ 1 & c & c^2 \end{vmatrix} \\ &= (a-b)(b-c)(c-a) \begin{vmatrix} 0 & -1 \\ 1 & b+c \end{vmatrix} \end{aligned}$$

$$\text{Hence, } \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = (a-b)(b-c)(c-a) \text{ proved.}$$

Conclusion

(ii) Let

$$\begin{aligned} \Delta &= \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix} \\ \Delta &= \begin{vmatrix} 0 & 0 & 1 \\ a-c & b-c & c \\ a^3-c^3 & b^3-c^3 & c^3 \end{vmatrix} \quad [C_1 \rightarrow C_1 - C_3 \text{ and } C_2 \rightarrow C_2 - C_3] \\ &= \begin{vmatrix} 0 & 0 & 1 \\ a-c & b-c & c \\ (a-c)(a^2+a c+c^2) & (b-c)(b^2+b c+c^2) & c^3 \end{vmatrix} \\ &= (c-a)(b-c) \begin{vmatrix} 0 & 0 & 1 \\ -1 & 1 & c \\ -(a^2+a c+c^2) & (b^2+b c+c^2) & c^3 \end{vmatrix} \end{aligned}$$

Applying $C_1 \rightarrow C_1 + C_2$,

$$\begin{aligned} \Delta &= (c-a)(b-c) \begin{vmatrix} 0 & 0 & 1 \\ 0 & 1 & c \\ (b^2-a^2)+(b c-a c) & (b^2+b c+c^2) & c^3 \end{vmatrix} \\ &= (b-c)(c-a)(a-b) \begin{vmatrix} 0 & 0 & 1 \\ 0 & 1 & c \\ -(a+b+c) & (b^2+b c+c^2) & c^3 \end{vmatrix} \\ &= (a-b)(b-c)(c-a)(a+b+c) \begin{vmatrix} 0 & 0 & 1 \\ 0 & 1 & c \\ -1 & (b^2+b c+c^2) & c^3 \end{vmatrix} \\ &= (a-b)(b-c)(c-a)(a+b+c)(-1) \begin{vmatrix} 0 & 1 \\ 1 & c \end{vmatrix} \\ &= (a-b)(b-c)(c-a)(a+b+c) \end{aligned}$$

$$\text{Hence, } \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(a+b+c) \text{ proved.}$$

Q

9

Properties of
determinants



By using properties of determinants show that:

$$\begin{vmatrix} x & x^2 & yz \\ y & y^2 & zx \\ z & z^2 & xy \end{vmatrix} = (x-y)(y-z)(z-x)(x+y+z)$$

Conclusion

Let

$$\Delta = \begin{vmatrix} x & x^2 & yz \\ y & y^2 & zx \\ z & z^2 & xy \end{vmatrix}$$

$$\begin{aligned}
\Delta &= \begin{vmatrix} x & x^2 & yz \\ y-x & y^2-x^2 & zxy-z \\ z-x & z^2-x^2 & xy-yz \end{vmatrix} [R_2 \rightarrow R_2 - R_1 \text{ and } R_3 \rightarrow R_3 - R_1] \\
&= \begin{vmatrix} x & x^2 & yz \\ -(x-y) & -(x-y)(x+y) & z(x-y) \\ (z-x) & (z-x)(z+x) & -y(z-x) \end{vmatrix} \\
&= (x-y)(z-x) \begin{vmatrix} x & x^2 & yz \\ -1 & -x-y & z \\ 1 & (z+x) & -y \end{vmatrix} \\
\Delta &= (x-y)(z-x) \begin{vmatrix} x & x^2 & yz \\ -1 & -x-y & z \\ 0 & z-y & z-y \end{vmatrix} \\
&= (x-y)(z-x)(z-y) \begin{vmatrix} x & x^2 & yz \\ -1 & -x-y & z \\ 0 & 1 & 1 \end{vmatrix} \\
&= [(x-y)(z-x)(z-y)] [(-1) \begin{vmatrix} x & yz \\ -1 & z \end{vmatrix} + 1 \begin{vmatrix} x & x^2 \\ -1 & -xy \end{vmatrix}] \\
&= (x-y)(z-x)(z-y) [(-x - yz) + (-x^2 - xy)] \\
&= -(x-y)(z-x)(z-y)(x + y + z) \\
&= (x-y)(y-z)(z-x)(x + y + z)
\end{aligned}$$

Hence, $\begin{vmatrix} x & x^2 & yz \\ y & y^2 & zx \\ z & z^2 & xy \end{vmatrix} = (x-y)(y-z)(z-x)(x + y + z)$ proved.

Q

10

Properties of
determinants



By using properties of determinants show that: (i)

$$\begin{vmatrix} x+4 & 2x & 2x \\ 2x & x+4 & 2x \\ 2x & 2x & x+4 \end{vmatrix} = (5x+4)(4-x)^2$$

(ii) $\begin{vmatrix} y+k & y & y \\ y & y+k & y \\ y & y & y+k \end{vmatrix} = k^2(3y+k)$

Given

(i)

$$\Delta = \begin{vmatrix} x+4 & 2x & 2x \\ 2x & x+4 & 2x \\ 2x & 2x & x+4 \end{vmatrix}$$

$$\Delta = \begin{vmatrix} 5x+4 & 5x+4 & 5x+4 \\ 2x & x+4 & 2x \\ 2x & 2x & x+4 \end{vmatrix} [R_1 \rightarrow R_1 + R_2 + R_3]$$

$$= (5x+4) \begin{vmatrix} 1 & 1 & 1 \\ 2x & x+4 & 2x \\ 2x & 2x & x+4 \end{vmatrix}$$

$$= (5x+4) \begin{vmatrix} 1 & 0 & 0 \\ 2x & -x+4 & 0 \\ 2x & 0 & -x+4 \end{vmatrix} [C_2 \rightarrow C_2 - C_1 \text{ and } C_3 \rightarrow C_3 - C_1]$$

$$= (5x+4)(4-x)(4-x) \begin{vmatrix} 1 & 0 & 0 \\ 2x & 1 & 0 \\ 2x & 0 & 1 \end{vmatrix}$$

$$= (5x+4)(4-x)^2 \begin{vmatrix} 1 & 0 \\ 2x & 1 \end{vmatrix}$$

$$= (5x+4)(4-x)^2$$

proved.

Conclusion

(ii)

$$\Delta = \begin{vmatrix} y+k & y & y \\ y & y+k & y \\ y & y & y+k \end{vmatrix}$$

$$\Delta = \begin{vmatrix} 3y+k & 3y+k & 3y+k \\ y & y+k & y \\ y & y & y+k \end{vmatrix} [R_1 \rightarrow R_1 + R_2 + R_3]$$

$$= (3y+k) \begin{vmatrix} 1 & 1 & 1 \\ y & y+k & y \\ y & y & y+k \end{vmatrix}$$

$$= (3y+k) \begin{vmatrix} 1 & 0 & 0 \\ y & k & 0 \\ y & 0 & k \end{vmatrix} [C_2 \rightarrow C_2 - C_1 \text{ and } C_3 \rightarrow C_3 - C_1]$$

$$= k^2(3y+k) \begin{vmatrix} 1 & 0 & 0 \\ y & 1 & 0 \\ y & 0 & 1 \end{vmatrix}$$

Expanding along C_3

$$\Delta = k^2(3y+k) \begin{vmatrix} 1 & 0 \\ y & 1 \end{vmatrix}$$

$$= k^2(3y+k)$$

$$\text{Hence, } \begin{vmatrix} y+k & y & y \\ y & y+k & y \\ y & y & y+k \end{vmatrix} = k^2(3y+k) \text{ proved.}$$

Q

11

Properties of
determinants

(())

By using properties of determinants show that: (i)

$$\begin{vmatrix} a-b-c & 2a & 2a \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix} = (a+b+c)^3 \text{ (ii)}$$

$$\begin{vmatrix} x+y+2z & x & y \\ z & y+z+2x & y \\ z & x & z+x+2y \end{vmatrix} = 2(x+y+z)^3$$

Given

$$(i) \Delta = \begin{vmatrix} a-b-c & 2a & 2a \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix}$$

$$\Delta = \begin{vmatrix} a+b+c & a+b+c & a+b+c \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix} [R_1 \rightarrow R_1 + R_2 + R_3]$$

$$= (a+b+c) \begin{vmatrix} 1 & 1 & 1 \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix}$$

$$= (a+b+c) \begin{vmatrix} 1 & 0 & 0 \\ 2b & -(a+b+c) & 0 \\ 2c & 0 & -(a+b+c) \end{vmatrix} [C_2 \rightarrow C_2 - C_1 \text{ and } C_3 \rightarrow C_3 - C_1]$$

$$= (a+b+c)^3 \begin{vmatrix} 1 & 0 & 0 \\ 2b & -1 & 0 \\ 2c & 0 & -1 \end{vmatrix}$$

$$= (a+b+c)^3 (-1)(-1)$$

$$= (a+b+c)^3$$

Hence, proved.

Conclusion

(ii)

$$\begin{aligned}
 \Delta &= \begin{vmatrix} x+y+2z & x & y \\ z & y+z+2x & y \\ z & x & z+x+2y \end{vmatrix} \\
 \Delta &= \begin{vmatrix} 2(x+y+z) & x & y \\ 2(x+y+z) & y+z+2x & y \\ 2(x+y+z) & x & z+x+2y \end{vmatrix} \quad [C_1 \rightarrow C_1+C_2+C_3] \\
 &= 2(x+y+z) \begin{vmatrix} 1 & x & y \\ 1 & y+z+2x & y \\ 1 & x & z+x+2y \end{vmatrix} \\
 &= 2(x+y+z) \begin{vmatrix} 1 & x & y \\ 0 & x+y+z & 0 \\ 0 & 0 & x+y+z \end{vmatrix} \quad [R_2 \rightarrow R_2-R_1 \text{ and } R_3 \rightarrow R_3-R_1] \\
 &= 2(x+y+z)^3 \begin{vmatrix} 1 & x & y \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} \\
 &= 2(x+y+z)^3 (1)(1-0) \\
 &= 2(x+y+z)^3
 \end{aligned}$$

Hence, proved.

Q

12

Properties of
determinants



By using properties of determinants show that:

$$\begin{vmatrix} 1 & x & x^2 \\ x^2 & 1 & x \\ x & x^2 & 1 \end{vmatrix} = (1-x^3)^2$$

Conclusion

$$\begin{aligned}
 \Delta &= \begin{vmatrix} 1 & x & x^2 \\ x^2 & 1 & x \\ x & x^2 & 1 \end{vmatrix} \\
 &= \begin{vmatrix} 1+x+x^2 & 1+x+x^2 & 1+x+x^2 \\ x^2 & 1 & x \\ x & x^2 & 1 \end{vmatrix} \quad [R_1 \rightarrow R_1+R_2+R_3] \\
 &= (1+x+x^2) \begin{vmatrix} 1 & 1 & 1 \\ x^2 & 1 & x \\ x & x^2 & 1 \end{vmatrix} \\
 \Delta &= (1+x+x^2) \begin{vmatrix} 1 & 0 & 0 \\ x^2 & 1-x^2 & x-x^2 \\ x & x^2-x & 1-x \end{vmatrix} \quad [C_2 \rightarrow C_2-C_1 \text{ and } C_3 \rightarrow C_3-C_1] \\
 \Delta &= (1+x+x^2)(1-x)(1-x) \begin{vmatrix} 1 & 0 & 0 \\ x^2 & 1+x & x \\ x & -x & 1 \end{vmatrix} \\
 &= (1-x^3)(1-x) \begin{vmatrix} 1 & 0 & 0 \\ x^2 & 1+x & x \\ x & -x & 1 \end{vmatrix}
 \end{aligned}$$

Expanding along R_1

$$\begin{aligned}
 \Delta &= (1-x^3)(1-x)(1) \begin{vmatrix} 1+x & x \\ -x & 1 \end{vmatrix} \\
 &= (1-x^3)(1-x)(1+x+x^2) \\
 &= (1-x^3)(1-x^3) \\
 &= (1-x^3)^2
 \end{aligned}$$

Hence, proved.

Q

13

Properties of
determinants

By using properties of determinants show that:

$$\begin{vmatrix} 1+a^2-b^2 & 2ab & -2b \\ 2ab & 1-a^2+b^2 & 2a \\ 2b & -2a & 1-a^2-b^2 \end{vmatrix} = (1+a^2+b^2)^3$$

Conclusion

$$\begin{aligned} \Delta &= \begin{vmatrix} 1+a^2-b^2 & 2ab & -2b \\ 2ab & 1-a^2+b^2 & 2a \\ 2b & -2a & 1-a^2-b^2 \end{vmatrix} \\ &= \begin{vmatrix} 1+a^2+b^2 & 0 & -b(1+a^2+b^2) \\ 0 & 1+a^2+b^2 & a(1+a^2+b^2) \\ 2b & -2a & 1-a^2-b^2 \end{vmatrix} \quad [R_1 \rightarrow R_1 + b R_3 \text{ and } R_2 \rightarrow R_2 - a R_3] \\ &= (1+a^2+b^2)^2 \begin{vmatrix} 1 & 0 & -b \\ 0 & 1 & a \\ 2b & -2a & 1-a^2-b^2 \end{vmatrix} \\ &= (1+a^2+b^2)^2 \left[(1) \begin{vmatrix} 1 & a \\ -2a & 1-a^2-b^2 \end{vmatrix} - b \begin{vmatrix} 0 & 1 \\ 2b & -2a \end{vmatrix} \right] \\ &= (1+a^2+b^2)^2 \left[1-a^2-b^2+2a^2-b(-2b) \right] \\ &= (1+a^2+b^2)^2 (1+a^2+b^2) \\ &= (1+a^2+b^2)^3 \end{aligned}$$

Hence, proved.

Q

14

Properties of
determinants

By using properties of determinants show that:

$$\begin{vmatrix} a^2+1 & ab & ac \\ ab & b^2+1 & bc \\ ca & cb & c^2+1 \end{vmatrix} = 1+a^2+b^2+c^2$$

Conclusion

$$\Delta = \begin{vmatrix} a^2+1 & ab & ac \\ ab & b^2+1 & bc \\ ca & cb & c^2+1 \end{vmatrix}$$

Taking out common factors a, b, c from R_1, R_2, R_3 respectively,

$$\begin{aligned}
 \Delta &= a \begin{vmatrix} a + \frac{1}{a} & b & c \\ a & b + \frac{1}{b} & c \\ a & b & c + \frac{1}{c} \end{vmatrix} \\
 &= a \begin{vmatrix} a + \frac{1}{a} & b & c \\ -\frac{1}{a} & \frac{1}{b} & 0 \\ -\frac{1}{a} & 0 & \frac{1}{c} \end{vmatrix} \quad [R_2 \rightarrow R_2 - R_1 \text{ and } R_3 \rightarrow R_3 - R_1] \\
 &= a \begin{vmatrix} a + \frac{1}{a} & b & c \\ -\frac{1}{a} & \frac{1}{b} & 0 \\ -\frac{1}{a} & 0 & \frac{1}{c} \end{vmatrix} \times \frac{1}{abc} \begin{vmatrix} a^2+1 & b^2 & c^2 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{vmatrix} \quad [C_1 \rightarrow a C_1, C_2 \rightarrow b C_2 \text{ and } C_3 \rightarrow c C_3] \\
 &= \begin{vmatrix} a^2+1 & b^2 & c^2 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{vmatrix} \\
 &= -1 \begin{vmatrix} b^2 & c^2 \\ 1 & 0 \end{vmatrix} + 1 \begin{vmatrix} a^2+1 & b^2 \\ -1 & 1 \end{vmatrix} \\
 &= -1(-c^2) + (a^2+1+b^2) \\
 &= 1+a^2+b^2+c^2
 \end{aligned}$$

Hence, proved.

Q

15

Properties of determinants

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Let A be a square matrix of order 3×3 , then $|kA|$ is equal to: (A) $k|A|$ (B) $k^2|A|$ (C) $k^3|A|$ (D) $3^k|A|$

Given

Let

$$A = \begin{pmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{pmatrix}$$

Then,

$$\begin{aligned}
 kA &= \begin{pmatrix} ka_1 & kb_1 & kc_1 \\ ka_2 & kb_2 & kc_2 \\ ka_3 & kb_3 & kc_3 \end{pmatrix} \\
 |kA| &= \begin{vmatrix} ka_1 & kb_1 & kc_1 \\ ka_2 & kb_2 & kc_2 \\ ka_3 & kb_3 & kc_3 \end{vmatrix}
 \end{aligned}$$

Taking out common factors k from each row

$$\begin{aligned}
 |kA| &= k^3 \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \\
 &= k^3 |A|
 \end{aligned}$$

Conclusion

The correct option is C.

Q

16

Properties of determinants



Which of the following is correct? (A) Determinant is a square matrix. (B) Determinant is a number associated to a matrix. (C) Determinant is a number associated to a square matrix. (D) None of the above.

Given

Every square matrix $A = [a_{ij}]$ of order n has an associated number.

This number is called the determinant of the square matrix A , where $a_{ij} = (i, j)^{\text{th}}$ element of A .

Conclusion

Thus, the determinant is a number associated to a square matrix.

Hence, the correct option is C.

Q

2

Evaluating a determinant without expanding



Without expanding the determinant, prove that

$$\begin{vmatrix} a & a^2 & bc \\ b & b^2 & ca \\ c & c^2 & ab \end{vmatrix} = \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix}.$$

Key formula

$$\begin{aligned} \text{LHS} &= \begin{vmatrix} a & a^2 & bc \\ b & b^2 & ca \\ c & c^2 & ab \end{vmatrix} \\ &= \frac{1}{abc} \begin{vmatrix} a^2 & a^3 & abc \\ b^2 & b^3 & abc \\ c^2 & c^3 & abc \end{vmatrix} \quad [R_1 \rightarrow aR_1, R_2 \rightarrow bR_2, R_3 \rightarrow cR_3] \\ &= \frac{1}{abc} \cdot abc \begin{vmatrix} a^2 & a^3 & 1 \\ b^2 & b^3 & 1 \\ c^2 & c^3 & 1 \end{vmatrix} \quad [\text{Taking out factor } abc \text{ from } C_3] \\ &= \begin{vmatrix} a^2 & a^3 & 1 \\ b^2 & b^3 & 1 \\ c^2 & c^3 & 1 \end{vmatrix} \\ &= \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix} \quad [C_1 \leftrightarrow C_3 \text{ and } C_2 \leftrightarrow C_3] \\ &= \text{RHS} \end{aligned}$$

Conclusion

Hence, proved.

Q

4

Determinant identity with $a+b+c$ symmetry

If a, b, c are real numbers and $\Delta = \begin{vmatrix} b+c & c+a & a+b \\ c+a & a+b & b+c \\ a+b & b+c & c+a \end{vmatrix} = 0$, show that either $a+b+c = 0$ or $a = b = c$.

Given

$$\begin{aligned}\Delta &= \begin{vmatrix} b+c & c+a & a+b \\ c+a & a+b & b+c \\ a+b & b+c & c+a \end{vmatrix} \\ &= \begin{vmatrix} 2(a+b+c) & 2(a+b+c) & 2(a+b+c) \\ c+a & a+b & b+c \\ a+b & b+c & c+a \end{vmatrix} [R_1 \rightarrow R_1 + R_2 + R_3] \\ &= 2(a+b+c) \begin{vmatrix} 1 & 1 & 1 \\ c+a & a+b & b+c \\ a+b & b+c & c+a \end{vmatrix} \\ &= 2(a+b+c) \begin{vmatrix} 1 & 0 & 0 \\ c+a & b-c & b-a \\ a+b & c-a & c-b \end{vmatrix} [C_2 \rightarrow C_2 - C_1 \text{ and } C_3 \rightarrow C_3 - C_1]\end{aligned}$$

Expanding R_1 ,

$$\begin{aligned}\Delta &= 2(a+b+c)(1)[(b-c)(c-b) - (b-a)(c-a)] \\ &= 2(a+b+c)[-b^2 - c^2 + 2b \cdot c - b \cdot c + b \cdot a + a \cdot c - a^2] \\ &= 2(a+b+c)[a \cdot b + b \cdot c + c \cdot a - a^2 - b^2 - c^2]\end{aligned}$$

Conclusion

It is given that $\Delta = 0$.

Hence,

$$2(a+b+c)[a \cdot b + b \cdot c + c \cdot a - a^2 - b^2 - c^2] = 0$$

$$\text{Either } (a+b+c) = 0 \text{ or } [a \cdot b + b \cdot c + c \cdot a - a^2 - b^2 - c^2] = 0$$

Now,

$$\begin{aligned}\Rightarrow a \cdot b + b \cdot c + c \cdot a - a^2 - b^2 - c^2 &= 0 \\ \Rightarrow -2a \cdot b - 2a \cdot c - 2c \cdot a + 2a^2 + 2b^2 + 2c^2 &= 0 \\ \Rightarrow (a-b)^2 + (b-c)^2 + (c-a)^2 &= 0 \\ \Rightarrow (a-b)^2 = (b-c)^2 = (c-a)^2 = 0 & \text{ [(a-b)^2, (b-c)^2, (c-a)^2 are non-negative]} \\ \Rightarrow (a-b) = (b-c) = (c-a) &= 0 \\ \Rightarrow a = b = c\end{aligned}$$

Hence, if $\Delta = 0$, then either $(a+b+c) = 0$ or $a = b = c$.

Q

5

Solving a
determinant
equation for
x

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Solve the equations $\begin{vmatrix} x+a & x & x \\ x & x+a & x \\ x & x & x+a \end{vmatrix} = 0, a \neq 0$.

Given

$$\begin{aligned}\Rightarrow \begin{vmatrix} x+a & x & x \\ x & x+a & x \\ x & x & x+a \end{vmatrix} &= 0 \\ \Rightarrow \begin{vmatrix} 3x+a & 3x+a & 3x+a \\ x & x+a & x \\ x & x & x+a \end{vmatrix} &= 0 [R_1 \rightarrow R_1 + R_2 + R_3] \\ \Rightarrow (3x+a) \begin{vmatrix} 1 & 1 & 1 \\ x & x+a & x \\ x & x & x+a \end{vmatrix} &= 0 \\ \Rightarrow (3x+a) \begin{vmatrix} 1 & 0 & 0 \\ x & a & 0 \\ x & 0 & a \end{vmatrix} &= 0 [C_2 \rightarrow C_2 - C_1 \text{ and } C_3 \rightarrow C_3 - C_1]\end{aligned}$$

Expanding along R_1 ,

$$\begin{aligned}\Rightarrow (3x+a)[1 \times a^2] &= 0 \\ \Rightarrow a^2(3x+a) &= 0\end{aligned}$$

Conclusion

Since $a \neq 0$

Therefore,

$$\Rightarrow 3x + a = 0$$

$$\Rightarrow x = -\frac{a}{3}$$

Q**6**

Proving a
determinant
identity



Prove that

$$\begin{vmatrix} a^2 & bc & a+c^2 \\ a^2+ab & b^2 & ac \\ ab & b^2+bc & c^2 \end{vmatrix} = 4a^2b^2c^2.$$

Given

$$\begin{aligned} \Delta &= \begin{vmatrix} a^2 & bc & a+c^2 \\ a^2+ab & b^2 & ac \\ ab & b^2+bc & c^2 \end{vmatrix} \\ &= a b c \begin{vmatrix} a & c & a+c \\ a+b & b & a \\ b & b+c & c \end{vmatrix} \quad [\text{Taking out common factors } a, b \text{ and } c \text{ from } C_1, C_2 \text{ and } C_3] \\ &= a b c \begin{vmatrix} a & c & a+c \\ b & b-c & -c \\ b-a & b & -a \end{vmatrix} \quad [R_2 \rightarrow R_2 - R_1 \text{ and } R_3 \rightarrow R_3 - R_1] \\ &= a b c \begin{vmatrix} a & c & a+c \\ a+b & b & a \\ b-a & b & -a \end{vmatrix} \quad [R_2 \rightarrow R_2 + R_1] \\ &= a b c \begin{vmatrix} a & c & a+c \\ a+b & b & a \\ 2b & 2b & 0 \end{vmatrix} \quad [R_3 \rightarrow R_3 + R_2] \\ &= 2 a b^2 c \begin{vmatrix} a & c & a+c \\ a+b & b & a \\ 1 & 1 & 0 \end{vmatrix} \\ \Delta &= 2 a b^2 c \begin{vmatrix} a & c-a & a+c \\ a+b & -a & a \\ 1 & 0 & 0 \end{vmatrix} \quad [C_2 \rightarrow C_2 - C_1] \end{aligned}$$

Conclusion

Expanding along R_3 ,

$$\begin{aligned} \Delta &= 2 a b^2 c [a(c-a) + a(a+c)] \\ &= 2 a b^2 c [a c - a^2 + a^2 + a c] \\ &= 2 a b^2 c (2 a c) \\ &= 4 a^2 b^2 c^2 \end{aligned}$$

Hence, proved.

Q**11**

Using
properties of
determinants
to prove an
identity



Using properties of determinants prove that:

$$\begin{vmatrix} \alpha & \alpha^2 & \beta+\gamma \\ \beta & \beta^2 & \gamma+\alpha \\ \gamma & \gamma^2 & \alpha+\beta \end{vmatrix} = (\beta-\gamma)(\gamma-\alpha)(\alpha-\beta)(\alpha+\beta+\gamma)$$

Key formula



$$\begin{aligned}
 \Delta &= \begin{vmatrix} \alpha & \alpha^2 & \beta+\gamma \\ \beta & \beta^2 & \gamma+\alpha \\ \gamma & \gamma^2 & \alpha+\beta \end{vmatrix} \\
 &= \begin{vmatrix} \alpha & \alpha^2 & \beta+\gamma \\ \beta-\alpha & \beta^2-\alpha^2 & \alpha-\beta \\ \gamma-\alpha & \gamma^2-\alpha^2 & \alpha-\gamma \end{vmatrix} \quad [R_2 \rightarrow R_2-R_1 \text{ and } R_3 \rightarrow R_3-R_1] \\
 &= (\beta-\alpha)(\gamma-\alpha) \begin{vmatrix} \alpha & \alpha^2 & \beta+\gamma \\ 1 & \beta+\alpha & -1 \\ 1 & \gamma+\alpha & -1 \end{vmatrix} \quad . = (\alpha-\beta)(\beta-\gamma)(\gamma-\alpha)(\alpha+\beta+\gamma), = \text{---} \\
 &= (\beta-\alpha)(\gamma-\alpha) \begin{vmatrix} \alpha & \alpha^2 & \beta+\gamma \\ 1 & \beta+\alpha & -1 \\ 0 & \gamma-\beta & 0 \end{vmatrix} \quad [R_3 \rightarrow R_3-R_2] \\
 &= (\beta-\alpha)(\gamma-\alpha)[-(\gamma-\beta)(-\alpha-\beta-\gamma)] \quad [\text{Expanding along } R_3] \\
 &= (\beta-\alpha)(\gamma-\alpha)(\gamma-\beta)(\alpha+\beta+\gamma)
 \end{aligned}$$

Conclusion

Hence, proved.

Q

12

Using properties of determinants to prove an identity



Using properties of determinants prove that:

$$\begin{vmatrix} x & x^2 & 1+px^3 \\ y & y^2 & 1+py^3 \\ z & z^2 & 1+pz^3 \end{vmatrix} = (1+pxyz)(x-y)(y-z)(z-x)$$

Key formula



$$\begin{aligned}
 \Delta &= \begin{vmatrix} x & x^2 & 1+px^3 \\ y & y^2 & 1+py^3 \\ z & z^2 & 1+pz^3 \end{vmatrix} \\
 \Delta &= \begin{vmatrix} x & x^2 & 1+px^3 \\ y-x & y^2-x^2 & p(y^3-x^3) \\ z-x & z^2-x^2 & p(z^3-x^3) \end{vmatrix} \quad [R_2 \rightarrow R_2-R_1 \text{ and } R_3 \rightarrow R_3-R_1] \\
 \Delta &= (y-x)(z-x) \begin{vmatrix} x & x^2 & 1+px^3 \\ 1 & y+x & p(y^2+x^2+xy) \\ 1 & z+x & p(z^2+x^2+xz) \end{vmatrix} \\
 \Delta &= (y-x)(z-x) \begin{vmatrix} x & x^2 & 1+px^3 \\ 1 & y+x & p(y^2+x^2+xy) \\ 0 & z-y & p(z-y)(x+y+z) \end{vmatrix} \quad [R_3 \rightarrow R_3-R_2] \\
 \Delta &= (y-x)(z-x)(z-y) \begin{vmatrix} x & x^2 & 1+px^3 \\ 1 & y+x & p(y^2+x^2+xy) \\ 0 & 1 & p(x+y+z) \end{vmatrix} \\
 \Delta &= (x-y)(z-y)(z-x)[(-1)(p)(xy^2+x^3+x^2y)+1+px^3+p(x+y+z)(xy)] \quad [\text{Expanding along } R_3] \\
 &= (x-y)(y-z)(z-x)[-px^2y^2-px^3-px^2y+1+px^3+px^2y+px^2y+pxyz] \\
 &= (x-y)(y-z)(z-x)(1+pxyz)
 \end{aligned}$$

Conclusion

Hence, proved.

Q

13

Using
properties of
determinants
to prove an
identity

()

Using properties of determinants prove that:

$$\begin{vmatrix} 3a & -a+b & -a+c \\ -b+a & 3b & -b+c \\ -c+a & -c+b & 3c \end{vmatrix} = 3(a+b+c)(a+b+c+c+a)$$

Key formula

$$\begin{aligned} \Delta &= \begin{vmatrix} 3a & -a+b & -a+c \\ -b+a & 3b & -b+c \\ -c+a & -c+b & 3c \end{vmatrix} \\ &= \begin{vmatrix} a+b+c & -a+b & \\ a+b+c & 3b & \\ a+b+c & -b+c & \end{vmatrix} [C_1 \rightarrow C_1+C_2+C_3] \\ &= (a+b+c) \begin{vmatrix} 1 & -a+b & -a+c \\ 1 & 3b & -b+c \\ 1 & -c+b & 3c \end{vmatrix} \\ &= (a+b+c) \begin{vmatrix} 1 & -a+b & -a+c \\ 0 & 2b+a & a-b \\ 0 & a-c & 2c+a \end{vmatrix} [R_2 \rightarrow R_2-R_1 \text{ and } R_3 \rightarrow R_3-R_1] \\ &= (a+b+c)[(2b+a)(2c+a)-(a-b)(a-c)] \text{ [Expanding along } C_1] \\ &= (a+b+c)[4bc+2ab+2ac+a^2-a^2+ac+ba-bc] \\ &= (a+b+c)(3ab+3bc+3ca) \\ &= 3(a+b+c)(a+b+c+c+a) \end{aligned}$$

Conclusion

Hence, proved.

Q

14

Using
properties of
determinants
to prove an
identity

✧

Using properties of determinants prove that:

$$\begin{vmatrix} 1 & 1+p & 1+p+q \\ 2 & 3+2p & 4+3p+2q \\ 3 & 6+3p & 10+6p+3q \end{vmatrix} = 1$$

Key formula

$$\begin{aligned} \Delta &= \begin{vmatrix} 1 & 1+p & 1+p+q \\ 2 & 3+2p & 4+3p+2q \\ 3 & 6+3p & 10+6p+3q \end{vmatrix} \\ &= \begin{vmatrix} 1 & 1+p & 1+p+q \\ 0 & 1 & 2+p \\ 0 & 3 & 7+3p \end{vmatrix} [R_2 \rightarrow R_2-2R_1 \text{ and } R_3 \rightarrow R_3-3R_1] \\ &= \begin{vmatrix} 1 & 1+p & 1+p+q \\ 0 & 1 & 2+p \\ 0 & 0 & 1 \end{vmatrix} [R_3 \rightarrow R_3-3R_2] \\ &= 1 \begin{vmatrix} 1 & 2+p \\ 0 & 1 \end{vmatrix} \text{ [Expanding along } C_1] \\ &= 1(1-0) = 1 \end{aligned}$$

Conclusion

Hence, proved.

Q

15

Using
properties of
determinants
to prove an
identity

✧

Using properties of determinants prove that:

$$\begin{vmatrix} \sin \alpha & \cos \alpha & \cos(\alpha+\delta) \\ \sin \beta & \cos \beta & \cos(\beta+\delta) \\ \sin \gamma & \cos \gamma & \cos(\gamma+\delta) \end{vmatrix} = 0$$

Key formula



$$\begin{aligned}\Delta &= \begin{vmatrix} \sin \alpha & \cos \alpha & \cos (\alpha+\delta) \\ \sin \beta & \cos \beta & \cos (\beta+\delta) \\ \sin \gamma & \cos \gamma & \cos (\gamma+\delta) \end{vmatrix} \\ &= \frac{1}{\sin \delta \cos \delta} \begin{vmatrix} \sin \alpha \sin \delta & \cos \alpha \cos \delta & \cos \alpha \cos \delta - \sin \alpha \sin \delta \\ \sin \beta \sin \delta & \cos \beta \cos \delta & \cos \beta \cos \delta - \sin \beta \sin \delta \\ \sin \gamma \sin \delta & \cos \gamma \cos \delta & \cos \gamma \cos \delta - \sin \gamma \sin \delta \end{vmatrix} \\ &= \frac{1}{\sin \delta \cos \delta} \begin{vmatrix} \cos \alpha \cos \delta & \cos \alpha \cos \delta & \cos \alpha \cos \delta - \sin \alpha \sin \delta \\ \cos \beta \cos \delta & \cos \beta \cos \delta & \cos \beta \cos \delta - \sin \beta \sin \delta \\ \cos \gamma \cos \delta & \cos \gamma \cos \delta & \cos \gamma \cos \delta - \sin \gamma \sin \delta \end{vmatrix} \quad [C_1 \rightarrow C_1 + C_3]\end{aligned}$$

Conclusion

Here, two columns C_1 and C_2 are identical.

Therefore, $\Delta = 0$

Hence, proved.

Q

17

Determinant
identity for
terms in A.P.



If a, b, c are in A.P, then the determinant

$$\begin{vmatrix} x+2 & x+3 & x+2a \\ x+3 & x+4 & x+2b \\ x+4 & x+5 & x+2c \end{vmatrix} \text{ is (A) 0 (B) 1 (C) x (D) 2x}$$

Key formula



$$\begin{aligned}\Delta &= \begin{vmatrix} x+2 & x+3 & x+2a \\ x+3 & x+4 & x+2b \\ x+4 & x+5 & x+2c \end{vmatrix} \\ &= \begin{vmatrix} x+2 & x+3 & x+2a \\ x+3 & x+4 & x+(a+c) \\ x+4 & x+5 & x+2c \end{vmatrix} \quad (2b = a+c \text{ as } a, b, c \text{ are in A.P}) \\ &= \begin{vmatrix} -1 & -1 & a-c \\ x+3 & x+4 & x+(a+c) \\ 1 & 1 & c-a \end{vmatrix} \quad [R_1 \rightarrow R_1 - R_2 \text{ and } R_3 \rightarrow R_3 - R_2] \\ &= \begin{vmatrix} 0 & 0 & 0 \\ x+3 & x+4 & x+a+c \\ 1 & 1 & c-a \end{vmatrix} \quad [R_1 \rightarrow R_1 + R_3]\end{aligned}$$

Conclusion

Here, all the elements of the first row are zero.

Hence, we have $\Delta = 0$

Thus, the correct option is A.